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# THE COLONIZATION AND CONSERVATION GENETICS OF THE RAINBOW DARTER, *ETHEOSTOMA CAERULEUM*, IN NORTHEAST IOWA

An Abstract of a Thesis

Submitted

in Partial Fulfillment

of the Requirements for the Degree

Master of Science

Dane Jonathan Davis

University of Northern Iowa

May 2012

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ABSTRACT

The aim of this project was to study the population genetic structure of the rainbow darter, *Etheostoma caeruleum*, distributed in northeastern Iowa rivers. Overall, Iowa sustains a wide array of physical habitats and varying fish assemblage conditions that are influenced by climate and the geology of the landscape. The chief objective of the research was to increase knowledge of nongame species movement that can aid managers in conservation.

This project addresses the question: What is the population genetic structure, amount of gene flow, and level of connectivity between populations of *Etheostoma caeruleum* distributed in northeast Iowa. Specimens were collected from rivers that are found in two geographically distinct regions in Iowa called the Iowan Surface and Paleozoic Plateau. Tissue samples were collected from 270 individuals that represent 14 localities. Microsatellite markers were amplified with PCR techniques, scored, and analyzed for population genetic structure. However, two of the ten microsatellite loci were unusable because of null alleles. Lack of detectable population subdivision coupled with excess heterozygotes from a bottleneck offer evidence that rainbow darters colonized northeast Iowa after the Wisconsinan episode. Even though these regions were circumnavigated by the Wisconsinan glacial advance, the Paleozoic Plateau and Iowan Surface did not act as a refuge for the rainbow darters. Historical processes were reflected in the rainbow darter's population genetic structure in northeast Iowa more than recently established anthropogenic barriers.

# THE COLONIZATION AND CONSERVATION GENETICS OF THE RAINBOW DARTER, *ETHEOSTOMA CAERULEUM*, IN NORTHEAST IOWA

A Thesis

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Dane Jonathan Davis

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This is a Study by: Dane Jonathan Davis

Entitled: Colonization and conservation genetics of the rainbow darter, *Etheostoma caeruleum,* in northeast Iowa

has been approved as meeting the thesis requirement for the

Degree of Master of Science

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# CHAPTER 1

# INTRODUCTION

The Quaternary glacial advances 2.6 million years ago (MYA) to 10,500 years ago (YA) had a profound effect on the current distribution of fish in the upper Mississippi River Region (Jacquemin and Pryon 2011). Glacial advances can force populations to relocate to southern refugia, place populations in allopatry, and alter genetic diversity via founder effect (Hewitt 2000). A product of glacial activity is reflected today by the presence of contrasting fish assemblages between glaciated and unglaciated regions. Another consequence of Quaternary glacial activity is sequestered gene flow between fish populations leading to speciation and divergence (Avise *et al.*1998). Along with historical processes, impoundments and low-head dams can also affect fish assemblage (Santucci Jr. *et al.* 2005), and foster genetic substructure in fish populations (Haponski *et al.* 2007).

Overall, Iowa sustains a wide array of physical habitats and varying fish assemblage conditions that are influenced by climate and the geology of the landscape (Heitke *et al.* 2006). The rainbow darter, *Etheostoma caeruleum*, is a specialist that prefers fast riffles (fast shallow currents in rivers) where it can feed on macroinvertebrates (Heithaus and Laushman 1997) and is considered an indicator species by the EPA to measure water quality (Tonnis 2006). This low-migratory fish inhabits northeast Iowa and is not found anywhere else in the state (Wilson 1987). Northeast Iowa contains two geographically distinct regions the Iowan Surface and Paleozoic Plateau. The Iowan Surface and Paleozoic Plateau have undergone glaciation during the

Pre-Illinoian advances and both regions were circumnavigated by the Illinoian and Wisconsinan glacial advances. The main component of the Iowan Surface includes slopes with low gradients that consequently lead to slower moving streams (Prior 1991). A combination of water, wind, and slope movements during periods of heavy erosion in the late Pleistocene impacted the topography of the Iowan Surface (Walters 2008). In stark contrast, the Paleozoic Plateau has steep, rugged terrain with faster moving streams, and a bedrock formation that controls the landscape dynamics (Prior 1991). The Paleozoic Plateau provides refuge for 40% of the endangered and threatened vertebrates found in Iowa (Howe 1984).

Haponski *et al.* (2009) studied the population genetic structure of the rainbow darter in Ohio using eight microsatellite loci and two mitochondrial (mt)DNA genes. They found *Etheostoma caeruleum* exhibited low-levels of migration and had little gene flow between populations. Ray *et al.* (2006) did a phylogeographic study of the entire range of the rainbow darter using the mitochondrial cytochrome *b* gene. *Etheostoma caeruleum* is comprised of four geographically distinct clades that includes the Eastern clade in western Pennsylvania, Ohio, Indiana, Illinois, Michigan, New York, West Virginia, Kentucky, Tennessee, and northern Alabama, the Northern Ozarks-upper Midwest clade in Wisconsin, northeast Iowa, Minnesota, and southern Missouri, the Mississippi River corridor clade in Arkansas, southern Missouri, and Mississippi, and the White-Little Red river clade in southern Missouri and Arkansas. The Mississippi River corridor and White-Little Red river clades are insular to unglaciated regions, whereas the Eastern and Northern Ozarks-upper Midwest clades encompasses glaciated and

unglaciated regions. They found cytochrome *b* diversity of *Etheostoma caeruleum* to be higher in unglaciated than glaciated regions.

The chief objective of the research was to increase knowledge of nongame species movement that can aid managers in conservation. This project addresses the questions: What is the population genetic structure, amount of gene flow, and level of connectivity between populations of *Etheostoma caeruleum* distributed in the Iowan Surface and Paleozoic Plateau? I hypothesize that, because *Etheostoma caeruleum* is a habitat specialist, it will show high population subdivision and low levels of migration. The landscape genetics concept serves a vital purpose in identifying how fish gene flow is influenced by a combination of anthropogenic structures, such as low-head dams and natural land structures formed by glacial activity, such as varying land topography. Haponski *et al.* (2009) sampled *Etheostoma caeruleum* in areas of Ohio glaciated by the Wisconsinan. However, no study with microsatellites has been conducted on *Etheostoma caeruleum* populations that were circumnavigated by the Wisconsinan advance or in the upper Mississippi River Basin. The expected outcome for the project includes accumulated results of the barriers between populations, which ultimately allow managers to formulate conservation units and locate population subdivisions. The finalized data set and analyses should contribute pertinent information for managers on the genetic integrity of the populations of these fish.

# CHAPTER 2

# LITERATURE REVIEW

## Quaternary Glacial Advances

Glacial activity that occurred throughout the Quaternary period, 2.6 MYA to 10,500 YA, contributed to the current landscape dynamics in Iowa (Prior 1991). A cooling climate in the Quaternary had a pronounced effect on creating conditions for glacial advances in North America and Europe (Bintanja and van de Wal 2008). The Quaternary glaciation episodes in the North American advance consisted of the Pre-Illinoian, Illinoian, and Wisconsinan episodes (Figure 1). The Pre-Illinoian episode consisted of multiple glacial advances lasting from 2.6 MYA to 500,000 YA (Bintanja and van de Wal 2008). These glacial advances extended as far south as northeast Kansas, northern Kentucky, and northern Missouri and all the way eastward to Massachusetts. The Illinoian glacial advance happened 300,000 to 130,000 YA and was less extensive than the Pre-Illinoian glacial advances. The Illinoian glacial advance east of the Mississippi River encroached upon southern areas of Illinois, Indiana, and Ohio. However, the glacial advance did not penetrate as far south in areas west of the Mississippi River when compared to glacial advances east of the Mississippi River. Only a small region of northwest Iowa was glaciated during the Illinoian advance and the glacial terminus ended in southeastern boundary of South Dakota.

The most recent glacial advance, the Wisconsinan, occurred 30,000 to 10,500 YA (Figure 1). The Laurentide ice sheet covered North America during the Wisconsinan episode and the Des Moines lobe is a southern extension of this ice sheet. The Des

Moines Lobe encroached into north central Iowa with the terminus ending where today the city of Des Moines, Iowa is located. The ice lobe was 200 km wide and 900 km long during its maximum extension southward from the main body of the Laurentide ice sheet (Patterson 1997). When the Des Moines lobe retreated, the signature of past glacial activity influence on the landscape includes glacial outwash from water causing sand and gravel to form stratified, moraines (collection of rocks and soil forming a hill) and glacial lakes (Miller *et al.* 2008).

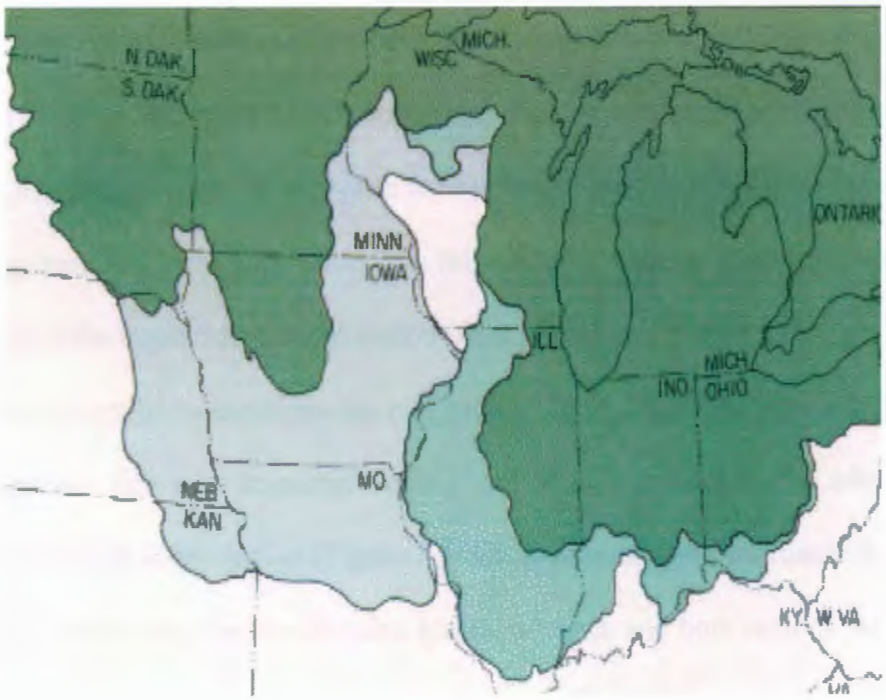


Figure 1. Quaternary Glacial Advances in the upper Midwest of North America. The Pre-Illinoian glacial advance 500,000 to 2,500,000 YA is shown in light grey. The light green color represents the Illinoian glacial advance 130,000 to 300,000 YA. The dark green color shows the farthest advance of the Wisconsin glacial advance 10,500 to 30,000 YA.
(http://www.igsb.uiowa.edu/Browse/glatrail/glatrail.htm retrieved February 11, 2012)

Glacial advances had a profound effect on the morphology of the Mississippi River over the past 100,000 years by influencing discharge and sediment load (Rittenour *et al.* 2007). Repeated glacial cycles in the upper Mississippi drainage basin impacted the landscape by the formation of terraces elevated above floodplains through downward erosion and caused valley degradation by the discharge of melt water (Bettis III *et al.* 2008). South of Cape Girardeau, Missouri marks a transitional boundary from Paleozoic bedrocks of the upper and middle Mississippi regions to more erodible sediments of the lower Mississippi River (Rittenour *et al.* 2007). As a result of increased sediment load and glacial melt discharge, the lower Mississippi River exhibited a braided drainage network pattern that lasted until the early Holocene 10,000 YA (Rittenour *et al.* 2007). Glaciers covering the upper Mississippi River region contributed to flooding and a higher flowing Mississippi River in the south fed by glacial melt water (Tripsanas *et al.* 2007). Frozen ground and loess deposition both influenced the rate of melt water and sediment discharge in the upper Mississippi region (Bettis III *et al.* 2008).

Northeast Iowa encompasses two geological regions, the Paleozoic Plateau and Iowan Surface, that were impacted directly or indirectly by past glacial advances in the upper Mississippi River region (Figure 2). These regions were encroached upon by glacial activity during the Pre-Illinoian glacial advance and both regions were by-passed by the Illinoian and Wisconsinan glacial advances (Figure 1). Further, a region located east of the Paleozoic Plateau, called the Driftless Area, was apparently never glaciated. The Driftless Area is located in southwest Wisconsin and northwest Illinois and

showcases bluffs up to 200 meters that were derived from stream and river erosion (Benke and Cushing 2005) (Figure 2).

Consequently, the extreme erosion during the Quaternary may explain the absence of glacial till within the Driftless Area (Iannicelli 2010). Solifluction, movement of wet earth material, and snowmelt sheetwash processes played a pronounced role in moving earth material and perhaps even extirpated evidence of glacial till in the Driftless Area (Iannicelli 2010). Mickelson et al. (1982) reviewed glacial evidence in the Driftless Area and concluded that no evidence of till does not prove that the Driftless Area was never glaciated. Conversely, there are hypotheses to explain why these regions were bypassed by glacial activity.

There are two hypotheses to explain why recent glacial advances did not penetrate the Driftless Area, Iowan Surface and Paleozoic Plateau. The first hypothesis proposes that sandstone in the region occluded glacial expansion in a process Hobbs (1999) described as dewatering of the base of the ice sheet. A second hypothesis by Cutler *et al.* (2001) suggests that the deep water of Lake Superior slowed the ice advance into the northern part of Wisconsin. In summation, factors such as a sieve, deep water, and perhaps ridges contributed to an ice-free Paleozoic Plateau and Iowan Surface during the Illinoian and Wisconsinan glacial periods.

## Iowan Surface (Iowa Erosional Surface)

The Iowan Surface, located in northeast Iowa and southeastern Minnesota displays long slopes that lack high relief and were formed by intense erosion during the late Quaternary period 15,000 YA (Koch 2004) (Figure 2). Consequently, the modern

topography of the Iowan Surface fosters conditions for poor drainage and streams with low gradients (Prior 1991). Modern drainage networks were probably established between the Wisconsinan and Illinoian glacial advances, which date to 125,000-75,000 YA (Vrecken 1975). The substrate under streams and rivers in this region tends to contain more gravel and coarse substrate when compared to other regions of Iowa (Rowe *et al.* 2009). In comparison with the streams and rivers of the Paleozoic Plateau, substrate within the Iowan Surface contains a higher concentration of sand and fewer substrates exceeding 16 mm in diameter (Rowe *et al.* 2009).

Figure 2. Map of the Geological Regions found in Iowa.
(http://uipress.lib.uiowa.edu/vpi/IowaFlora.aspx retrieved February 11, 2011).

Northeast Iowa was tundra during the Wisconsinan episode. The climate in northeast Iowa between 21,000 and 16,500 YA contributed to a landscape that featured tundra with permafrost (Walters 2008). Temperatures were 14°C cooler than Holocene temperatures following the retreat of Laurentide ice sheet during the Wisconsinan glacial period (Walters 1994). Evidence from the fossil record and unique landscape formations indicate that tundra once existed in Iowa. Fossils of plants, insects and small mammals associated with tundra environment were found in Conklin Quarry in southeast Iowa and were aged at 18,090-16,710 YA (Baker *et al.* 1986). Conifer forests were common in Iowa and the rest of North America during the Wisconsinan glacial episode as shown by pollen evidence (Jackson *et al.* 2000). A period of lower carbon dioxide concentration in the Wisconsinan era ushered in conditions well suited for coniferous forests to thrive (Jackson *et al.* 2000). The vegetation dynamics shifted from conifer to deciduous forests 10,000 YA then to prairie 5,500 YA (Bettis III and Autin 1997).

Additional evidence supporting past tundra conditions in northeast Iowa are inactive ice-wedge casts. An ice-wedge cast forms in periglacial conditions as layers of ice were added to a cast over time. Interestingly, inactive ice-wedge casts are left over relics from tundra conditions in northeast Iowa, which are comprised of silt, silty sand, and gravelly sand within the cast (Walters 1994). Ice-wedge casts are underlain by sediment transported by wind and sheet erosion, which supports the hypothesis that intense erosion occurred in the Iowan Surface (Walters 1994).

Water, wind, and slope movements all contributed to step-like landscape formations during periods of intense erosion in the Iowan Surface (Walters 2008).

Erosion began 29,000 YA and lasted until 18,000 YA when loess, windblown glacial sediment, deposition occurred (Vrecken 1975). The deposition of loess concluded around 14,000 YA. Areas of the Iowan Surface that were not covered by loess between 20,300 and 18,300 YA served as a source of sand, which was transported by wind onto loess terrain (Vrecken 1975). Along with wind sculpting the landscpae, the movement of water increased the amount of sediment in Eastern Iowa rivers and streams during the conclusion of the Wisconsinan episode (Bettis III and Autin 1997). The efficiency of water to transport sediment was influenced by shifts in vegetation composition and a warmer Holocene climate 10,000 YA that can manipulate the amount of runoff and flood frequencies (Bettis III and Autin 1997).

Other land formations along the Iowan Surface reflect both glacial and wind activity in sediment composition. For example, a paha is a type of hill found on the Iowan Surface that is made up of Pre-Illinoian till and paleosol, a type of sediment blanketed by loess (Walters 2008). Sand and silt transported by the wind coats the surface of the paha (Walters 2008). Along with the paha formation, stringers and ventifacts formations offer evidence of strong wind activity in the Iowan Surface. Stringers, which stand three to five meters above the surface, in a northwest-southwest orientation indicates a more prominent northwest wind pattern (Koch 2004). Ventifacts are a type of rock that forms pits by wind blowing particles into it and is usually distributed along a stoneline (Davidson 2010). Ventifacts were formed approximately 21,000 to 16,500 YA during periods of intense erosion and offer further evidence of

strong winds (Davidson 2010). These land formations formed by winds offer clues about the condition of the environment that species lived in during the Wisconsinan period.

Geographical evidence that champions the hypothesis of heavy erosion in the Iowan Surface during the Wisconsinan episode includes the presence of stonelines above a layer of Pre-Illinoian till and Wisconsinian alluvium occurrences (Splinter *et al.* 2005). The region underwent natural erosion cycles during the Wisconsinan and now human activity plays the primary role in shaping the topography. Humans began to have a more pronounced role in moving sediment approximately 1,000 YA, which has led to natural erosion being replaced with erosion induced by humans as a primary force in sediment movement (Wilkinson 2005). The present Iowan Surface displays human driven impacts caused by long-term agriculture, which has lead to slope erosion and accumulation of sediment along valley bottoms (Vrecken 1975).

## Paleozoic Plateau

The Paleozoic Plateau encompasses northeastern Iowa, northwestern Illinois, southeastern Minnesota, and southwestern Wisconsin (Figure 2). A Silurian Escarpment marks the border between the Iowan Surface and Paleozoic Plateau that has varying solubility of earth material in each region (Prior 1991). In stark contrast to the Iowan Surface geography, the Paleozoic Plateau features hill prairies with rocky soils that cover the underlying limestone bedrock in northeast Iowa (Schennum 1986). The hill prairies are very steep with exposures oriented in a South and West direction (Schennum 1986). Consequently, the steep gradient of streams entrenched in bedrock contribute to fast

moving streams and rivers (Prior 1991). Ground water discharge causes the river drainage networks to have stable and cool temperatures (Rowe *et al.* 2009).

The dominant substrates found in Paleozoic Plateau river networks are gravel and coarse substrate (Rowe *et al.* 2009). The higher percentage of coarse substrate found in northern Iowa tributaries are a left over product of glacial activity (Menzel 1987). River anatomy affects fish assemblages, as does the amount of vegetation. Components of the anatomy of a river that can promote or discourage fish species diversity include the percent of substrate composition, channel morphology, and depth (Rowe *et al.* 2009). The lack of sand and fine substrate in these rivers may encourage the assemblage of specialized fish that prefer gravel over sand (Rowe *et al.* 2009). The extensive canopy cover in this region is comprised of indigenous maples, *Acer* spp., and basswood, *Tilia* spp., which offer shade for rivers and streams (Rowe *et al.* 2009). Sites rich in canopy cover in Iowa Rivers displayed a higher diversity of fish when compared to tributaries lacking canopy cover (Rowe *et al.* 2009).

## River Morphology in Northeast Iowa

Rivers that flow through the Iowan Surface and Paleozoic Plateau in northeast Iowa include the Cedar, Upper Iowa, Turkey, Volga, Yellow, Wapsipinicon and Maquoketa rivers (Figure 3). These rivers are all tributaries of the upper Mississippi River. Glacial activity had a pronounced effect on watersheds by establishing new drainage networks and increasing water flow. The Cedar River and higher elevated Wapsipinicon River both flow through the Iowan Surface; however, glacial impact was more pronounced in the Cedar River than in the Wapsipinicon River. Quaternary glacial

advances left a major imprint on the Cedar River with a larger channel width, less frequent side channels and more discharge when compared to the Wapsipinicon River (Lee 2008). Interestingly, Fields (2008) suggested that bedrock topography offers evidence that the Cedar River and Wapsipinicon River were once connected.

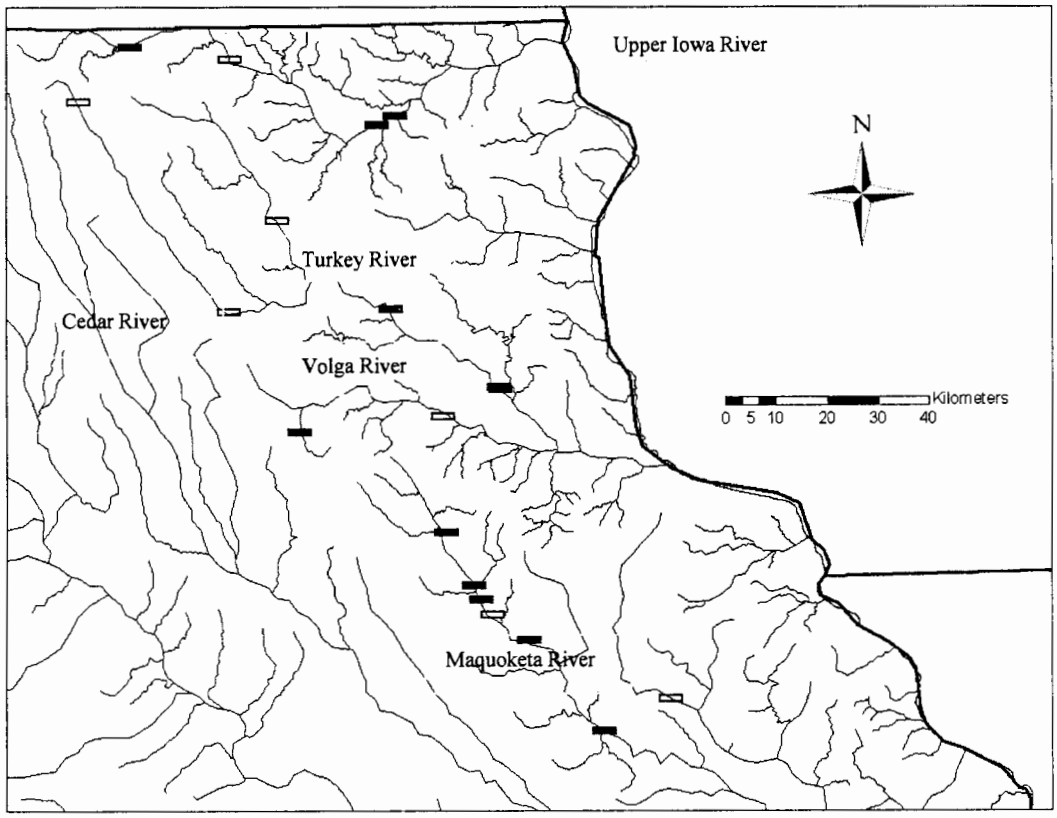


Figure 3. Rivers of Northeast Iowa and Dam distribution. Solid black lines indicate dams greater than 1.22 meters. Open lines indicate dams less than 1.22 meters. Map by Jack Kosmicki, University of Northern Iowa.

River networks keep evolving over time in regards to connectivity to other streams, shape, and flow. As the Des Moines Lobe of the Wisconsinan glacial maximum,

retreated north in Iowa, the Cedar River water volume shrank in size as a result of less glacial water feeding the system (Walters 2004). Subsequently, a proliferation of sediment load occurred within these systems during the late Wisconsinan as a result of loess accumulation and erosion (Bettis III and Autin 1997). The glacial water that once contributed to the formation of a larger Cedar River played a role in eroding the Cedar River Valley (Walters 2004). Not only are rivers and streams formed by glacial activity, lakes in the upper Mississippi region have also have glacial origins (Thorson 2009).

## The Impact of Glacial Activity on the Biota of Northeast Iowa

Shifts of species range in response to glacial activity can ultimately lead to contrasting genetic richness between regions. During a glacial advance, a species may either persist in an unglaciated pocket within a northern region or retreat south for more suitable habitat. Regions that were not covered by ice can even act as a rufugium assuming the microclimate remains constant to prevent species expiration (Stewart and Lister 2001). A second potential response to a glacial advance is that a species can no longer survive in an unglaciated pocket and is forced to relocate to new areas (Hewitt 1999). As a melting glacier retreats north, species can potentially colonize new territories that were previously glaciated (Hewitt 1999). The colonizing population tends to have less genetic diversity and greater homozygosity than the southern refuge source population (Hewitt 1996). A southern refuge features more genetic richness and population substructure when compared to a founding population (Avise 2000).

Populations of organisms can undergo hybridization, divergence, and speciation as a result of glacial activity influencing their spatial distribution (Hewitt 1996). Should

a population remain intact in a northern refugium, then hybridization may occur with the expanding southern population (Hewitt 1996). However, few hybrid zones were able to persist in North America because of climate oscillation (Barton and Hewitt 1985). Sometimes populations do not interact with each other to hybridize and may be placed into allopatry (Bilton *et al.* 1998). Consequently, glacial advances forcing populations into allopatry may lead to speciation (Bilton *et al.* 1998).

Regions of the upper Mississippi River yielded two outcomes in species distribution during glacial activity by acting either as a refugium and even forcing populations to contract their range south of the glacial advance. During Quaternary glacial episodes, northern regions by-passed by glacial activity acted as refugia for organisms (Howe 1984). Small mammals such as the eastern chipmunk, *Tamlas striatus*, were able to tolerate cooler temperatures in northern refugia in the upper Mississippi region (Rowe *et al.* 2004). Amphibians were even able to flourish in northern pockets of Wisconsin and northeast Iowa. Wood frogs in a northern refugium in Wisconsin expanded their range to Michigan and Ontario as the Wisconsinan advance retreated north (Lee-Yaw *et al.* 2008). The blue-spotted salamander, *Ambystoma laterale*, might have colonized new habitat from a refuge in northeast Iowa along with several other refugia distributed across the northeastern section of North America (Demastes *et al.* 2007). However, there is a possibility that the blue-spotted salamander may have colonized new territories by expanding its range northward, e.g. New England, and then westwards towards the Great Lakes from a single, eastern refugium (Demastes *et al.*

2007). Nonetheless, *Ambystoma laterale* was able to tolerate periglacial conditions in northern refugia along with eastern chipmunks and wood frogs.

Other organisms such as the Ozark minnow, *Notropis nubilus*, were unable to thrive in periglacial conditions of northern pockets and were temporally forced into southern refugia. The Ozark minnow, *Notropis nubilus*, encountered harsh conditions in the Paleozoic Plateau, which ultimately lead to expiration (Berendzen *et al.* 2010). After the Wisconsinan advance retreated, the Ozark minnows recolonized the Paleozoic Plateau from a refugium in the Ozark Highlands (Berendzen *et al.* 2010). Consequently, the genetic diversity of Ozark minnow of Paleozoic Plateau would be lower than organisms that persisted in northern pockets.

Currently, the Paleozoic Plateau is an area of high diversity but also contains 40% of Iowa's endangered and threatened vertebrates (Howe 1984). The Pleistocene snail, *Discus macclintocki*, is an example of an endangered species that inhabits algific slopes in this region (Ross 1999). Despite the abundance of imperiled species found in the Paleozoic Plateau, this region maintains a high Index of Biological Integrity (IBI). The IBI value measures the degree of impairment human activities have on water quality in streams and rivers. Compared to other regions in Iowa, the impact of human activity on rivers and streams were less pronounced in northeast Iowa (Heitke *et al.* 2006). The lower volume of human impact on northeast Iowa tributaries is reflected by a higher fish IBI than other regions in Western and Southern Iowa (Heitke *et al.* 2006).

## Impoundments and Low-head Dams

Not only did geological processes like glacial events have an impact on fish distribution, but the recent introduction of anthropogenic barriers also have a pronounced influence. Impoundments and low-head dams, dams that allow water over the top, can potentially impact fish abundance and richness by manipulating habitat dynamics (Santucci Jr. *et al.* 2005). Impoundments disrupt aquatic ecosystems by making the habitat more homogeneous (Santucci Jr. *et al.* 2005). A consequence of a reduced variety of water velocities includes an increase in silt and sand substrates (Santucci Jr. *et al.* 2005). An unaltered river system includes varying velocity rates, different water depths and variable substrate types and displays greater fish abundance and richness (Santucci Jr. *et al.* 2005).

Implications of dams on aquatic ecosystems include impeding fish migration, altering the nutrient cycle, and contributing a shift in habitat configuration (Santucci Jr. *et al.* 2005). The impact a dam has on the ability of fish to migrate depends mostly on the design of the dam. For instance, federal dams with a by-pass system in place allow salmon to migrate easily along the Snake River in the Pacific Northwest (Levin and Tolimieri 2001). On the other hand, private dams lacking the infrastructure to facilitate salmon migration were shown to negatively impact migration (Levin and Tolimieri 2001). There are only 42 high quality free flowing streams of 200 km or greater remaining in the lower 48 states (Benke 1990). The majority of the 5.2 million km of rivers that flow through the lower 48 have been altered by human activity (Benke 1990).

In Iowa, the Cedar River averages a dam every 15 miles, which amounts to 20 dams distributed along the channel (Fields 2008).

Even small anthropogenic barriers can impact the gene flow among fish populations. For example, the presence of mill weirs, which is a type of structure that creates ponds by damming small streams, along the Viaur River in Europe caused a decrease of gene flow, migration, and heterozygosity of freshwater fish in this river (Blanchet *et al.* 2010). Analyses of microsatellite data revealed that the smallest fish, the European minnow, *Phoxinus phoxinus*, was least affected by fragmentation from mill weirs (Blanchet *et al.* 2010). On the other hand, medium sized fish such as the rostrum dace, *Leuciscus leuciscus*, and gudgeon, *Gobio gobio*, were the most impacted by mill weirs. The Greek chub, *Leuciscus cephalus*, was impacted intermediately by mill weirs (Blanchet *et al.* 2010). In summation, the size of the fish dictated the degree of impact weir mills had on gene flow, migration, and heterozygosity.

Low-head dams located in Fox River, Illinois were found to negatively impact warm water fish by jeopardizing habitat integrity (Santucci Jr. *et al.* 2005). Impoundment of the Pond River system in Kentucky may restrict the ability of the Shawnee darter, *Etheostoma tecumehi*, to migrate downstream and ultimately hinders gene flow between fragmented habitats (Cicerello and Butler 2007). A study of the greenside darter, *Etheostoma blennioides,* and logperch darter, *Percina caprodes*, found that *P. caprodes* located below a low-head dam had unique haplotypes suggesting a lack of gene flow across the barrier (Haponski *et al.* 2007). However, a shared haplotype in *E. blennioides* populations upstream and downstream of the barrier implied some gene flow. In

conclusion, Haponski *et al.* (2007) could not determine if the low-head dam was responsible for impeding gene flow or not.

Not all studies give credence to the role of a barrier in manipulating genetic structure of fish populations and providing certain size of fish an advantage in migration. For instance, one population of creek chub, *Semolitus aromaculatus,* separated by a manmade barrier, displayed similar genetic diversity as populations that lacked a barrier (Boizard *et al.* 2009). Haponski *et al.* (2007) found low-head dams might not separate populations of nonmigratory fish; however, the results were constrained by a small sample size. Along with a lack of concordance on the connection between dams and genetic diversity, fish morphology studies also yielded contradictory results. The size and shape of 48 species of fish representing 34 genera were not to be influenced by sea lamprey barriers (McLaughlin *et al.* 2006). McLaughlin *et al.* (2006) discredit swimming morphology and taxonomic affiliation as a means to gauge the sensitivity of a fish to barriers.

## Rainbow Darter, *Etheostoma caeruleum*

### Geographic Range and Physical Description

The rainbow darter, *Etheostoma caeruleum*, is distributed throughout rivers in central and eastern North America (Lee *et al.*1981) (Figure 4). However, the distribution of *E. caeruleum* in Iowa is limited to the northeastern portion of the state including the Paleozoic Plateau and Iowan Surface ecoregions (Figure 2). The rainbow darter is a small fish that can reach 7.62 cm in length (Wilson 1987) and belongs to the approximately 200 species of darters that inhabit eastern North America in the family

Percidae (Ray *et al.* 2008). Diagnostic features of *E. caeruleum* include six to ten vertical bars on each side (Marko 2003) and sexually dimorphic color pattern (Mendelson 2003) (Figure 5). Female rainbow darters have distinct brown stripes on its sides, whereas the male has blue stripes (Reeves 1907). A large adult male tends to be more colorful in the breeding season when compared to other seasons (Reeves 1907). The anal fin of the male rainbow darter contains shades of red, orange, and blue (Reeves 1907). In stark contrast, the female rainbow darter mostly maintains the same color pattern year round (Reeves 1907). During the breeding season, the first dorsal fin of a female rainbow darter undergoes a minute change in coloration (Reeves 1907). Based on estimates from otoliths, inner ear, these fish can live up to a maximum of five years (Beckman 2002).

*Etheostoma caeruleum* have a stout body and deep caudal peduncle, which enables them to live in high current areas with cobbles (Guill *et al.* 2003). Species of darters found in less extreme currents such as the Johnny darter, *Etheostoma nigrum*, and Iowa darter, *Etheostoma exile*, have narrow caudal peduncles (Guill *et al.* 2003). Along with the morphological advantages of being well suited for fast currents, certain morphological features also allow the rainbow darter to hunt efficiently. A small body enables the rainbow darter to feed more efficiently in tiny crevices (Hlohowskyj and Wissing 1986). A subterminal mouth enables it to effectively hunt prey located on rock surfaces (Paine *et al.* 1982). Due to poor lateral vision, rainbow darters are more efficient at attacking prey with a forward movement (Vogt Jr. and Coon 1990).

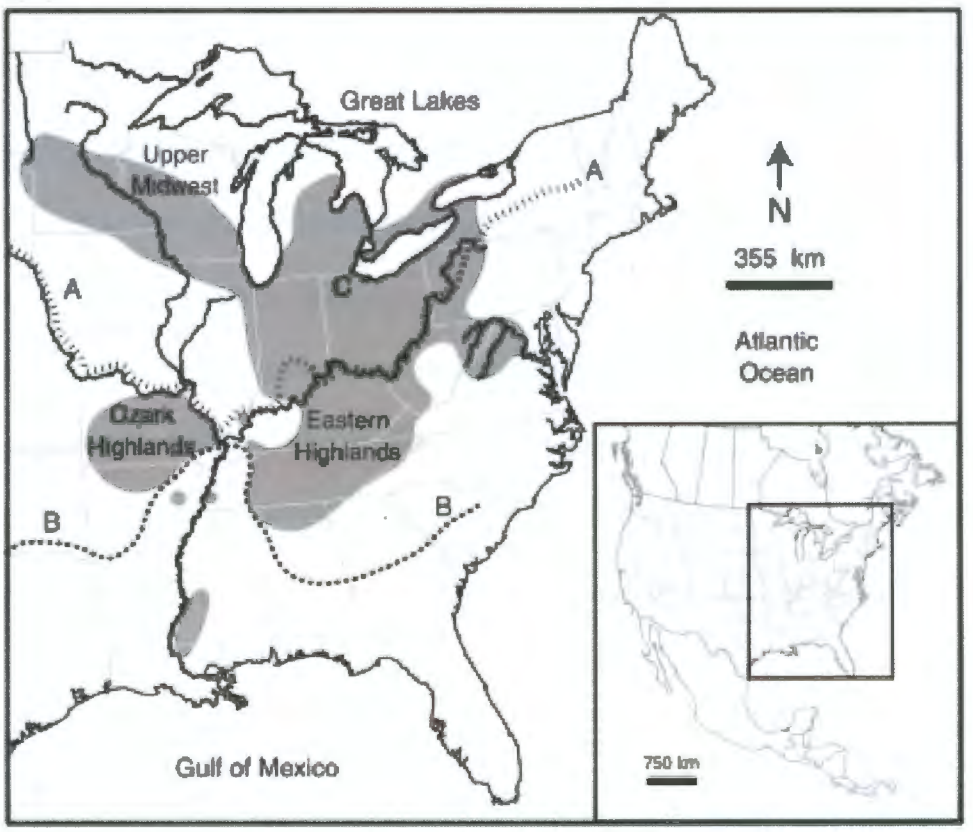


Figure 4. Distribution of *Etheostoma caeruleum* in North America. Grey indicates the range of the rainbow darter. The black solid line (A) indicates the farthest reach of the Pleistocene glacial advance, while the dotted line (B) indicates a former coastline. Figure from Ray *et al*. (2006).

Figure 5. A Male Rainbow Darter. Photo by Dr. Peter Berendzen

Feeding and Habitat

The rainbow darter is a specialist that prefers riffles with cobbles and gravel, and areas with fast shallow currents in rivers (Heithaus and Laushman 1997). On occasion they venture into slow moving pools for food (Vogt Jr. and Coon 1990). Pools act as an ideal hunting ground because of the low velocity to avert wasting energy and finer sediment bottoms that promote more widespread prey (Vogt Jr. and Coon 1990). Prime feeding times are in the daytime with two major peaks in feeding activity during early morning and late afternoon (Adamson and Wissing 1977). The diet of the rainbow darter comprises of on macroinvertebrates such as cadis fly larvae, e.g., *Hydroptilidae* larvae, Diptera larvae, e.g., *Tanytarsini* and *Orthocladiinae* (Paine *et al.* 1982) and occasionally small fish eggs (Adamson and Wissing 1977).

A predator of the rainbow darter, which is also found in riffles, is the mottled sculpin, *Cottus bairdi* (Harding *et al.* 1998). *Etheostoma caeruleum* respond to predators with alarm cues that induce stillness so the darter can blend in better with the environment (Crane *et al.* 2009). The rainbow darter is normally not a gregarious fish that forms schools; instead, most of its time is spent in solitude (Webb 1980).

For a rainbow darter to persist in a habitat, the habitat must meet a specific type of microhabitat and maintain good water quality. Microhabitat velocity was found to be a key factor in rainbow darter habitat selection in order to avoid wasting energy in extreme currents (Harding *et al.*1998). Furthermore, they were found to prefer water velocities that had a mean threshold of <3-5 cm $s^{-1}$ for microhabitats (Harding *et al.* 1998). Besides physical attributes of a habitat, water chemistry can determine if rainbow darters can exist

in a certain habitat. For example, a petroleum discharge contributed to the expiration of rainbow darters in Deleware Run, Ohio (Gatz Jr. and Haric 1993). Due to the sensitivity of the rainbow darter to pollution, the Environmental Protection Agency uses it as an indicator species to measure watershed integrity (Tonnis 2006).

Spawning

*Etheostoma caeruleum* spawn from April through June (Reeves 1907). The male competes for mates and the female constructs pit-like nests for depositing eggs (Reeves 1907). Once the female has ovulated, she creates a depression for laying eggs by digging her nose in the gravel bottom (Fuller 2003). The male mounts the top of the female and releases sperm into the depression. A female can spawn 39 times throughout the breeding season and deposit eight eggs per spawn (Fuller 1998a). The female rainbow darter can lay between 180-607 eggs with a mean annual fecundity of 309 eggs (Fuller 1998a). The average ripe egg size is 1.51 mm (Heins *et al.* 1996). The timeframe for the eggs to hatch once fertilized is at least 10 days (Marko 2003) where they stay attached to a stone within the pit (Reeves 1907).

In general, sexual selection plays a major role in driving the evolution of darters (Page and Swofford 1984). Typically male *Etheostoma caeruleum* occupies the breeding ground more frequently than females (Reeves 1907). Male rainbow darters tend to release more sperm in the presence of several males; however, larger males were found to postpone spawning when sperm competition predominates (Fuller 1998b). Furthermore, larger males tend to be guarders in spawning groups more often than smaller males (Fuller 1999). Large males do abandon the nest occasionally to look for more females

and drive away sneaky males but the large male eventually returns to guard the nest (Reeves 1907). Larger males get the opportunity to breed with a female more frequently; a dominant male can have higher fitness with 100% paternity passed down in a single spawning event (Fuller 1999). Aggression by the male guarder tends to influence reproductive fitness more so than the choosiness of a female in mate selection (Fuller 2003).

Population Genetic Structure

Quaternary glacial and interglacial episodes had a profound effect on the rainbow darter distribution and population genetic structure (Haponski *et al.* 2009; Ray *et al.* 2006). Haponski *et al.* (2009) studied populations of *E. caeruleum* from Lake Erie catchment of northern Ohio and Ohio River catchment of southern Ohio using eight microsatellite loci developed by Tonnis (2006) and two mtDNA genes. The six localities included in the study are areas covered by ice during Quaternary glacial advances. The main conclusion drawn by Haponski *et al.* (2009) was that *Etheostoma caeruleum* has a limited ability to migrate and lacked connectivity between populations. *Etheostoma caeruleum* in the Lake Erie catchment, which possessed less mtDNA diversity compared to Ohio River catchment, was able to access western Lake Erie by utilizing the Wabash and Maumee Rivers as a corridor (Haponski *et al.* 2009). The Ohio River catchment may have had greater mtDNA diversity because of its proximity to a source population. Genetic evidence of low migration and gene flow included no mtDNA haplotypes shared between catchments and each catchment had 25 and 31 private microsatellite alleles. The lack of mtDNA diversity in the Lake Erie catchment population suggests a past

bottleneck and a recent colonization. No difference was found in microsatellite diversity between the Lake Erie and Ohio River catchments (Haponski *et al.* 2009).

The phylogeographic study of rainbow darters by Ray *et al.* (2006) show limited gene flow between populations and the impact of glacial activity on the current genetic diversity. Based on analyses of the mtDNA cytochrome *b* gene, Ray *et al.* (2006) found four distinct clades of rainbow darters in the Mississippi drainage network with three of the clades located west of the Mississippi River (Figure 6). The White-Little Red river clade, which is found west of the Mississippi River and isolated to unglaciated regions, includes southern Missouri and northern Arkansas whereas the Eastern Clade only occurs east of the Mississippi River. The third clade, Mississippi River corridor, is found in unglaciated tributaries on both sides of the Mississippi River along the middle and lower reaches of the Mississippi River. The rainbow darters in northeast Iowa belong to the Northern Ozarks-upper Midwest clade that encompasses Iowa, Minnesota, Wisconsin and an insular population in Missouri. Interestingly the refugium in southern Missouri harbors three major clades of rainbow darters and the clade absent from this region is the Eastern clade. As glaciers melted and retreated north, new habitat became available and newly established drainage networks allowed *Etheostoma caeruleum* to colonize previously glaciated areas from refugia in southern Missouri and unglaciated regions east of the Mississippi River.

The present generation of rainbow darters exhibits low levels of migration and contrasting levels of genetic diversity between glaciated and unglaciated regions (Ray *et al.* 2006). Many populations have exclusive haplotypes and do not share many

haplotypes across drainages suggest low connectivity between populations. Only one haplotype was shared among *Etheostoma caeruleum* populations between glaciated and unglaciated areas (Ray *et al.* 2006). Populations inhabiting the recently glaciated area showed less genetic diversity as compared to unglaciated regions. Cytochrome *b* nucleotide diversity for *Etheostoma caeruleum* populations in the Northern Ozarks-upper Midwest clade was 0.017±0.009; while the unglaciated regions south of the Quaternary glacial maximum were richer with 0.035±0.020 nucleotide diversity. Based on the results of Ray *et al.* (2008) and Haponski *et al.* (2009), rainbow darter population genetics suggest low migration between populations, populations inhabiting recently glaciated areas contain lower genetic diversity than unglaciated areas, and the spatial distribution of rainbow darters was influenced by glacial activity establishing new drainage networks.

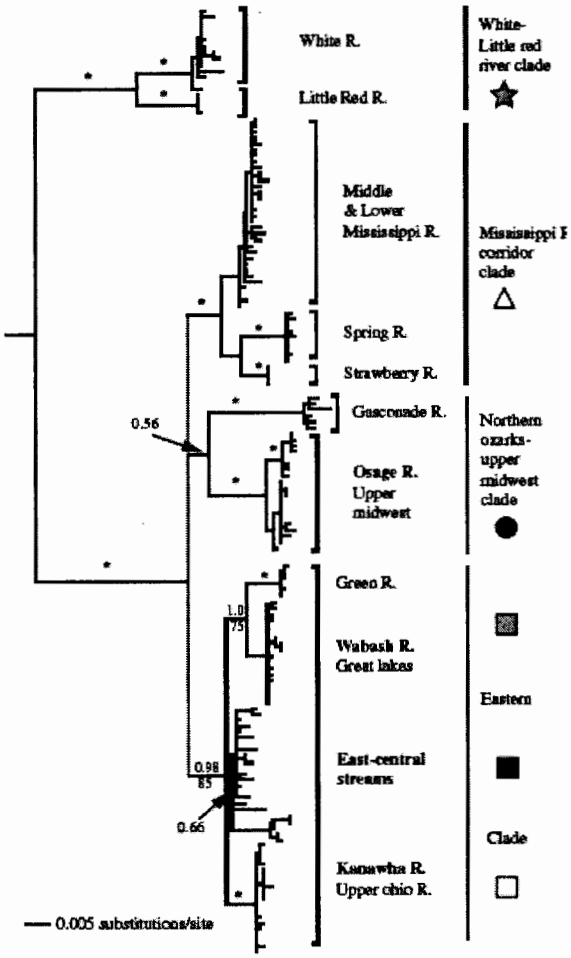


Figure 6. Phylogeny of *Etheostoma caeruleum* based on the mtDNA Cytochrome *b* Gene. Astricks represent bootstrap and posterior probabilities greater than 90%. Figure from Ray *et al.* (2006).

Not only can historical processes contribute to population genetic structure, but the life history of *Etheostoma caeruleum* also plays a vital role in gene flow (Turner and Trexler 1998). Life history factors that contribute more so to *Etheostoma caeruleum* gene flow influence on dispersal include egg size, female size, and fecundity (Turner and Trexler 1998). Natural selection favors large egg sizes and low fecundity should

resources become scarce (Bart and Page 1992). Large eggs produce larger offspring that can hunt benthic organisms sooner than offspring hatched from smaller eggs (Paine 1984). Darters exhibiting low migration rates, such as the rainbow darter, shared the characteristics of having small clutch sizes and large eggs (Turner and Trexler 1998). Consequently, low gene flow can lead to unique morphological differences between populations of rainbow darters (Turner and Trexler 1998).

## Conservation Genetics

Preserving and maintaining the genetic integrity of populations is a relevant component of conservation. Considering that the earth is experiencing the sixth greatest mass extinction (Foreman 2004), applying a conservation genetics approach to population management can alleviate this alarming trend. Two main objectives of conservation genetics are to foster genetic diversity and to avert inbreeding within populations (Hedrick and Miller 1992). A long-term advantage of high genetic diversity is the ability for a population to adapt and evolve in an environment that is prone to change (Meffe 1986). Genetically diverse populations tend to be rich in heterozygous individuals, which indicate less inbreeding and a higher level of fitness (Allendorf and Leary 1986). In stark contrast to large populations, small populations tend to have lower level of genetic variation and a reduction in population fitness (Hedrick 2001).

A population abated in size for at least a single generation can potentially undergo a bottleneck. Excess inbreeding from a bottleneck event can lower the fitness of a population by decreasing genetic variation (Hedrick and Miller 1992) and enabling genetic drift to play a more prominent role in fixation of alleles in small populations

(Ellstrand and Elam 1993). Bottleneck events have a direct connection to species range shifts resulting from glacial activity. A population colonizing a formerly glaciated region may experience a bottleneck with a loss of alleles along with a more homogenized population (Hewitt 1996). For example, a specific locus can experience a loss of allelic diversity through homogenization (Hewitt 1996). A short term bottleneck tends to not impact heterozygosity in a population as much as a population experiencing a long term bottleneck that may jeopardize its potential to recover (Allendorf 1986). The number of alleles left over following a bottleneck event may dictate the success of a population or species to persist in its habitat (Allendorf 1986).

Alleles can be introduced into a population by promoting gene flow, introducing individuals to a population, and mutation over time. A corridor is designed to promote suitable migration conditions and allow high survival rate between fragmented (Ferreras 2001; Mazerolle 2004). Establishing a corridor can facilitate gene flow between populations and introduce new alleles to a population (Shirk *et al.* 2010). A second avenue to add new alleles is by humans transporting and introducing individuals to a fragmented population. Individuals can be introduced to alleviate inbreeding within a population; however, introducing new individuals to a small population poses the risk of losing a beneficial allele by outbreeding depression (Hedrick 2001). Outbreeding depression lowers the fitness of an offspring if it shares parents from different populations. Alleles can also be introduced into a population without human intervention. For example, cheetahs experienced a bottleneck 12,000 YA that

homogenized the population, but new alleles were added through mutation (Driscoll *et al.* 2002).

Gene flow information can aid managers interested in the effectiveness corridors have in introducing new alleles to a population through fragmented landscapes (Shirk *et al.* 2010). The two types of fragmented habitats distributed across the modern landscape consist of areas fragmented by human activity (Pavlacky *et al.* 2009) and natural barriers (Wang 2009). Recently introduced anthropogenic barriers and natural barriers have been shown to limit gene flow between populations (Pavlacky *et al.* 2009; Wang 2009). Species were able to colonize new territories in North America as glaciers receded northward; however, a species ability to colonize new habitat in response to future global warming is limited by fragmentation of the landscape by human activity (Foreman 2004). Consequently, landscape has a profound role in the genetic makeup of populations and directly impacts how they are managed.

Landscape Genetics

Landscape genetics incorporates landscape ecology with population genetics to address how a landscape can manipulate genetic variation within or between populations (Manel *et al.* 2003). Barriers across a given landscape may impede gene flow depending on the quantity and quality of space that divides populations (Holderegger and Wagner 2008). For example, habitat loss or fragmentation can influence gene flow between populations (Holderegger and Wagner 2008). Other variables that impact the permeability between habitats include the behavior and movement capability of organisms (Cushman *et al.* 2006). Landscape genetics looks for potential relationships

between specific environmental features and genetic discontinuities within a common area (Manel *et al.* 2003). Testing landscape genetic hypotheses is accomplished by analyzing both genetic and landscape data (Holderegger and Wagner 2006).

Landscape genetic studies can be applied to questions on the impact of anthropogenic and natural barriers on gene flow among populations. A study of the population genetic structure of the Australian macropodid, *Macropus fuliginosus*, using nine microsatellite loci showed that mainland barriers such as the Swan River and Flinders Mountain Range impeded gene flow between populations (Neaves *et al.* 2009). Landscape barriers may also foster conditions for speciation to occur. A study of black frogs, *Bufo exsul*, in the Inyo-White Mountains of California found that a desert environment and limited dispersal ability contributed to divergence within group (Wang 2009). The impact of human activity on landscape connectivity was demonstrated by log runners, *Orthonyx temminckii*, with the birds reduced ability to disperse between populations due to fire suppression and homogenization of the landscape (Pavlacky *et al.* 2009). An underlying variable influencing gene flow in wood frogs, *Rana sylvatica*, was changes in landscape connectivity during and after European settlement (Zellmer and Knowles 2009). Analysis of nine microsatellite loci representing 51 populations of wood frogs suggested that the genetics reflected current landscape patterns more so than historical landscape patterns. These landscape genetic studies highlight how natural and human induced barriers can potentially dictate the genetic structure between populations. Consequently, barriers can contribute to speciation by allopatry, facilitate the formation

of genetically unique populations, and even contribute to inbreeding in a secluded population.

Microsatellites

Microsatellite loci are neutral markers from noncoding regions (Queller *et al.* 1993) that have a variable number of repeats along a sequence of DNA (Slatkin 1995). The tandem repeats can consist of dinucleotide, trinucleotide, or tetranucleotide repeats along a DNA sequence. Microsatellites undergo mutations by polymerase slippage at DNA replication (Jarne and Lagoda 1996), and the original allele size may dictate the size of the newly mutated allele (Slatkin 1995). A mutation can either result in the addition or deletion of a tandem repeat along the sequence. Because microsatellites are highly polymorphic, they are useful in studying population genetic structure (Slatkin 1995). Factors that influence the degree of polymorphism include population history, mating practices, and the abundance of individuals in the breeding population (Gerber *et al.* 2000).

Microsatellite markers can be used to determine effective population size, kinship and inbreeding occurrences within populations (Xu and Fu 2004). Effective population size ($N_e$) assumes that the size of the idealize population would show the same amount of genetic drift as the real population (Wright 1931). Variables that can influence $N_e$ include the sex ratio in a population, population subdivision, volatile changes in population size over generations, and mating practices (Caballero 1994). Hauser *et al.* (2002) estimated $N_e$ from microsatellite data found in New Zealand snapper *Pagrus auratus*. They found a decrease in heterozygotes and overall genetic diversity with a low

$N_e/N$ value, which is a ratio of effective population size to actual population size. Microsatellite markers are useful in kinship studies by distinguishing the relatedness of individuals to each other (Queller *et al.* 1993). Since microsatellites possess high levels of codominant polymorphism, this feature enables more accurate classification of parentage (Gerber *et al.* 2000). Lastly, colonization patterns and population subdivision in response to glacial activity can be estimated with microsatellite markers (Hewitt 2000).

These neutral markers do contain weaknesses when applied to other studies. Incorporating microsatellites into phylogenetic studies to test questions about the distant past can potentially yield a fallacious tree (Jarne and Lagoda 1996). This is due to high mutation rates in microsatellites contributing to homoplasy in phylogenetic trees. Because of the potential for homoplasy, nuclear and mitochondrial DNA are often preferred over microsatellites to construct phylogenetic trees (Zink and Barrowclough 2008). A second drawback to microsatellites is the onerous and arduous task of finding the actual marker (Queller *et al.* 1993).

# CHAPTER 3

# OBJECTIVES

The aim of this project was to determine the population genetic structure of the rainbow darter in northeast Iowa. These low migratory fish are distributed in two geographically distinct ecoregions in northeast Iowa, the Iowan Surface and Paleozoic Plateau. These regions were uniquely impacted by Quaternary glacial activity. The main objectives were to test for level of connectivity between populations and to define population subdivisions. Landscape components that can influence population genetic structure are anthropogenic and natural barriers. To determine how these barriers can potentially influence population dynamics, microsatellite markers were employed to test these objectives. Stemming from previous studies by Haponski *et al.* (2009) and Ray *et al.* (2006), I hypothesize that the rainbow darter in northeast Iowa will exhibit high population subdivision and low levels of migration. However, support of an alternative hypothesis would indicate recent colonization by the lack of population genetic structure or recent gene flow (Hewitt 2000).

# CHAPTER 4

# MATERIALS AND METHODS

## Site Description

Rainbow darters were collected from rivers that flow through two geographically distinct ecoregions in northeast Iowa, the Iowan Surface and Paleozoic Plateau. *Etheostoma caeruleum* were collected from the Cedar, Upper Iowa, Turkey, Volga and Maquoketa rivers (Figure 7; Table 1). All of these systems share the same characteristic of ultimately flowing into the Mississippi River. The sampled rivers that flow through the Paleozoic Plateau include the Upper Iowa, Turkey, and Volga rivers. The Cedar, Turkey, and Maquoketa rivers flow through the Iowan Surface. The Iowa Department of Natural Resources documented that rainbow darters have been collected in the Wapsipinicon River. However, attempts to collect rainbow darters at the Wapsipinicon River were unsuccessful. Efforts were made to collect sample above and below dams and impoundments.

Fish were collected by a combination of electroshocking and seining. Fin clips of specimens (fin-clip or entire fish) were deposited in 95% ethanol. Scissors were used to cut a piece of the caudal or pectoral fin. The scissors were sterilized between every fin-clip. A fin-clip was the primary method employed because it is a less invasive technique to abate fish morality. Clipped fish where released into the stream unharmed where they were obtained. Necessary permits from the Iowa Department of Natural Resources were acquired.

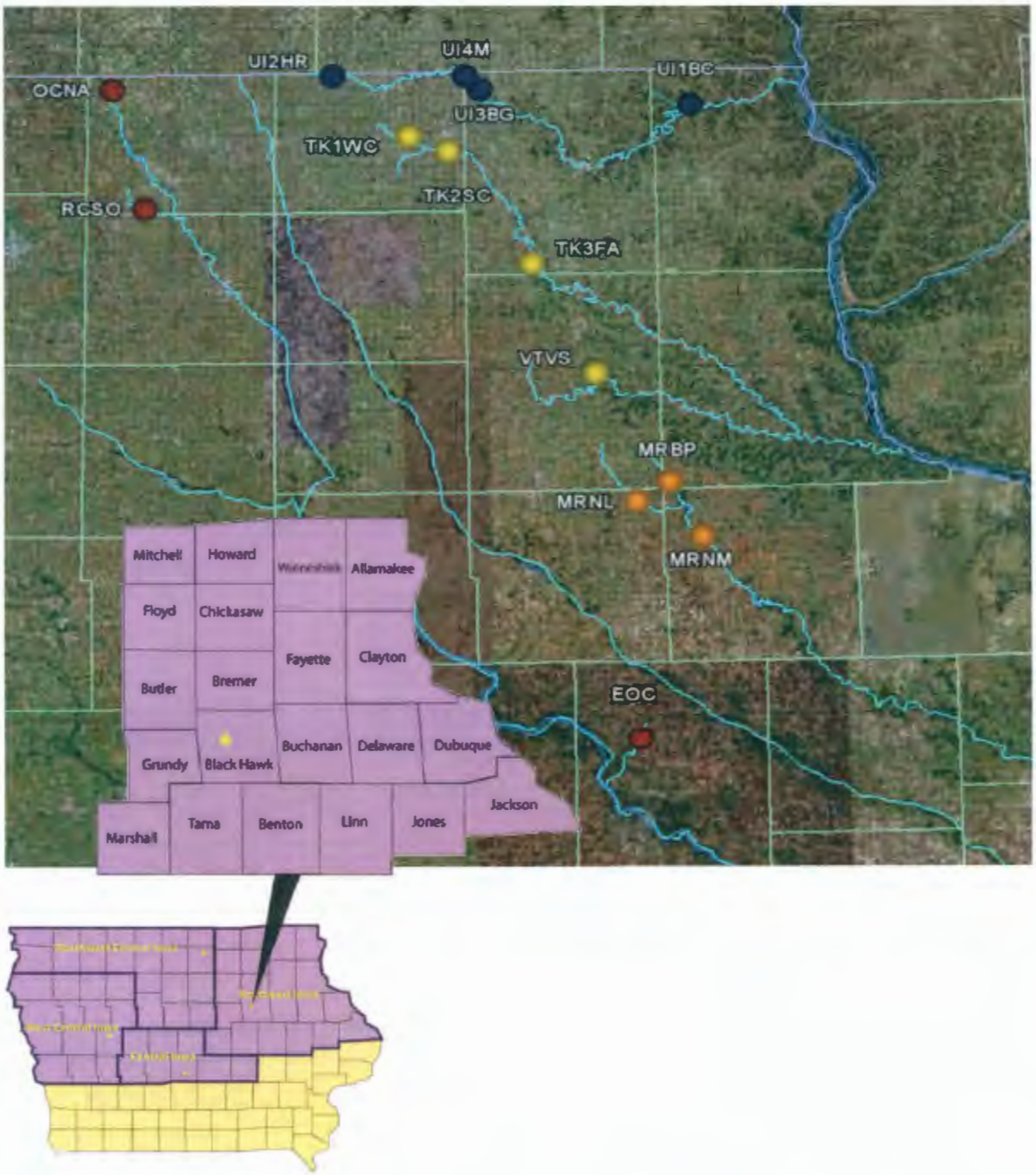


Figure 7. Map of Localities. The colored dots represents the following drainage: purple= Upper Iowa River, yellow=Turkey and Volga rivers, orange= Maquoketa River, and red= Cedar River. Table 1 offers the site identification (ID).

Table 1. Collection Localities of Rainbow Darter in Northeast Iowa. Collection sites denoted by region, locality identification (ID), total number of individuals collected from each locality (n) and general locality information. For the following drainages: UI=Upper Iowa River, TR=Turkey River, V=Volga, MR=Maquoketa River, and CR=Cedar River.

| Region/ID | n | County (IA) | Drainage | Lat (N)/Long (W) |
|---|---|---|---|---|
| **Paleozoic Plateau** | | | | |
| UI1BC | 10 | ALAMAKEE | UI | 43.421666/-91.553500 |
| UI2HR | 8 | HOWARD | UI | 43.493479/-92.405952 |
| UI3BG | 20 | WINNESHEIK | UI | 43.140280/-91.915550 |
| UI4M | 21 | WINNESHEIK | UI | 43.490790/-92.074530 |
| TK1WC | 30 | HOWARD | TR | 43.368110/-92.216790 |
| TK2SC | 30 | HOWARD | TR | 43.337150/-92.119710 |
| TK3FA | 25 | WINNESHEIK | TR | 43.493479/-91.405952 |
| **Iowan Surface** | | | | |
| VTVS | 12 | FAYETTE | V | 42.879481/-91.763618 |
| MRBP | 11 | CLAYTON | M | 42.658610/-91.587500 |
| MRNM | 15 | DELAWARE | M | 42.547870/-91.510150 |
| MRNL | 25 | BUCHANAN | M | 42.619910/-91.666950 |
| OCNA | 30 | MITCHELL | CR | 43.469588/-92.960773 |
| RCSO | 12 | MITCHELL | CR | 43.223680/-92.876650 |
| EOC | 22 | LINN | CR | 42.142530/-91.672300 |

## DNA Extraction

A standard guanidine thiocyanate extraction protocol was used to extract genomic DNA. A total of 300 ul of cell lysis buffer (1.975 M NaCl mM Tris-CL, 25mMEDTA, 5% SDS) was placed into each tube. Approximately 10 mg of the tissue was placed into the cell lysis buffer. A total of 2 ul Proteinase K (20mg/ml) was added to each tube and allowed to incubate overnight at 55°C. The reagents for DNA isolation were added in the following order: 1.5 ul RNAse (20mg/ml), 100 ul of protein precipitation buffer, 300 ul chilled isopropyl, and 300 ul 70% ethanol. Tubes were left with the cap off overnight to

desiccate completely. On the following day, 50 ul of Tris-Cl (10mM/4.2pH) was added and left overnight to incubate at room temperature. The stock solution of genomic DNA was then diluted using 10 ul of stock and 90 ul of nuclease free water. Both the stock and diluted DNA extracts were stored at 4°C.

Ten published microsatellite markers from Tonnis (2006) were amplified with PCR techniques. A single-reaction nested PCR method incorporating amplified and fluorescent labels in a single tube was carried out for loci that were shown to be scorable and polymorphic (Schuelke 2000; Boutin-Ganache *et al.* 2001). This PCR method was executed with three primers (Tonnis 2006) (Table 2): a forward M13 primer (1 mM) designed for the specific locus, reverse primer (10 mM) designed for the specific locus, and a M13 fluorescent primer (10 mM). The universal florescent-labeled M13 primers used for amplifications were HEX and FAM. A 25 ul cocktail that consisted of 1.5 ul of each primer, 9.0 ul of nuclease free water, 1 ul of DNA and 13.3 ul of GoTaq Hot Start Taq (Promega Corp. Madison, WI, USA) was amplified in a Bio-Rad MyCycler thermocycler. The following parameters were set for all PCR amplifications: initial denaturation at 94°C (2 minutes); 10 cycles of 94° C (20 seconds), 50°C (20 seconds), 72°C (30 seconds); 30 cycles of 94°C (20 seconds), 48°C (20 seconds), 72°C (30 seconds), and a final extension at 72°C (10 minutes). The PCR products were held at 4°C when the run was terminated. Amplifications were screened using 1.8% agrose gel electrophoresis for successful amplification. Amplified products were diluted with either 15 ul or 30 ul of nuclease free water for genotyping. The sizing of the diluted amplified

products was performed on an Applied Biosystems 3100 Genetic Analyzer at Iowa State University DNA Facility.

Table 2. Forward and Reverse Primer Sequences for the Microsatellite Loci used in the Study. Sequences from Tonnis (2006).

| Locus name | Primer | Sequence |
|---|---|---|
| ECA 6 EPA | F | 5'TGGTATGTGTGCGTGGATGTCATA3' |
| | R | 5'CTTTGAGCAAATCCCCTTACGGAA3' |
| ECA 10 EPA | F | 5'TGCACAATGAAGTTAAGATGCTGT3' |
| | R | 5'GAATGGCTCTGATATTGCATGATT3' |
| ECA 14 EPA | F | 5'CTACTTAAAACCCATGTTAGGTAA3' |
| | R | 5'CAGGGTCAGTGCTGGAAACA3' |
| ECA 22 EPA | F | 5'CAAGGGCAACTTTTTGGCTACT3' |
| | R | 5'GGTATAGAATCGTTACATTTTCTGAA3' |
| ECA 24 EPA | F | 5'AGTTTGCCCCAGAAATATGGAAA3' |
| | R | 5'GTCTGTGCTGTTTGACTGATTTT3' |
| ECA 36 EPA | F | 5'AGAAGCACTGAAGTATGGTGTT3' |
| | R | 5'GCTGCTTCAAATCTGCCAAATG3' |
| ECA 44 EPA | F | 5'AATGTTGCTGACGCAGATTGTA3' |
| | R | 5'ACTGGGACCATGAATTTCCA3' |
| ECA 46 EPA | F | 5'CTAAGCATGGTTTGGTTTGTGA3' |
| | R | 5'CCTTTTTTCCAGTGTCAGTGTCATTT3' |
| ECA 48 EPA | F | 5'AGCCATTGGTATGACCTTCAT3' |
| | R | 5'CACCAGCATCTTCCAACAGA3' |
| ECA 71 EPA | F | 5'CAATCCCTTCCTTCCAACCT3 |
| | R | 5'CATTTGCCAGTGACATCAAAA3' |

## Allele Scoring

Alleles were scored using GENEMARKER (version 1.85; SoftGenetics, State College, PA). Parameters for GENEMARKER were set to reflect the tetranucleotide repeat. MICROCHECKER v2.2.3 served as a quality assurance for GENEMARKER

scoring in order to check for large allele drop out, null alleles, and scoring errors from stutter peaks (van Oosterhout *et al.* 2004). The ten loci were analyzed using MICROCHECKER to check for any potential scoring errors in the population data set. Parameters were set at 1000 replicates and 95% confidence interval for each locus in MICROCHECKER.

Hardy-Weinberg equilibrium is the null hypothesis of evolution that assumes the absence of migration, mutations, and natural selection and random mating maintain constant allele frequencies over generations. A deviation from Hardy-Weinberg frequencies may indicate problems in genotyping (Wigginton *et al.* 2005). A marker not conforming to Hardy-Weinberg equilibrium can potentially lead to false conclusions (Wigginton *et al.*2005). The exact test of deviation from Hardy-Weinberg equilibrium for all loci was performed using GENEPOP version 4.1 (Raymond and Rousset 1995). Parameters were set for 10,000 dememorisations, 20 batches and 5,000 iterations under one population and the four drainages independently. Tests for Hardy-Weinberg equilibrium were also carried out in ARLEQUIN version 3.1 (Excoffier *et al.* 2005). Even though ARLEQUIN is analogous to GENEPOP with employing an exact test, these models contrast from each other by how the exact test calculates Hardy-Weinberg equilibrium. Parameters for ARLEQUIN were set at 1,000,000 steps in Markov chain and 100,000 dememorisation steps.

ARLEQUIN and GENEPOP were employed to check for linkage disequilibrium (LD) for a single large population including all 14 localities and by assigning localities to each specific drainage. Linked loci can be in linkage equilibrium and unlinked loci can

be in linkage disequilibrium from nonrandom association of genes at two loci. Factors that can contribute to high linkage disequilibrium are gene flow between genetically distinct populations (admixture), genetic drift following a bottleneck event, linkage in regards to the physical connection between loci, and selection (Flint-Garcia *et al.* 2003). Most genetic software programs require the genetic data to be in Hardy-Weinberg equilibrium and unlinked loci to achieve accurate results. Parameters for ARLEQUIN were set at 10,000 permutations and 0.05 as a statistically significant level to test for linkage disequilibrium. A likelihood ratio test as outlined in Slatkin and Excoffier (1996) was used to test for significant levels of linkage in ARLEQUIN. GENEPOP parameters were set at 1,000,000 dememorizations, 20 batches, and 100,000 iterations per batch. The log likelihood ratio statistic test (G-test) from Weir (1996) was used to test for significant levels of linkage.

Genetic Diversity and Population Genetic Structure

Wright (1921, 1969) established a method of testing for a deficiency of heterozygotes within and among populations through F-statistics. F-statistics encompass $F_{st}$, $F_{it}$, and $F_{is}$, and is useful in addressing questions in regards to population genetics. The lowercase letters in F-statistics represent the following: individual (i), subpopulations (s), and overall population (t). The equation of $F_{st}$ is $F_{st} = var(p)/p(1-p)$. The numerator contains the variance of p across the subpopulations, while the denominator has the expected frequency of heterozygotes. The $F_{st}$ value offers insight to the level of divergence between populations. A value close to 0 suggests low divergence; whereas a value closer to 1.0 indicates divergence and high population subdivision. The equation of

$F_{it}$ is $(1-F_{is})(1-F_{st})=(1-F_{it})$. $F_{it}$ measures the impact of inbreeding on heterozygosity in a total population. $F_{st}$ and $F_{it}$ were measured in GENEPOP. Pairwise $F_{st}$'s were calculated using a standard ANOVA as described by Weir and Cockerham (1984).

The number of alleles, allelic richness, and $F_{is}$ were quantified in FSTAT (Goudet 1995, 2001). Allelic richness accounts for population size influencing genetic diversity by measuring the number of alleles across all loci and the number of alleles in each locus. $F_{is}$ measures the amount of heterozygosity in a population with a positive number showing excess of homozygotes and conversely negative numbers showing excess heterozygotes. $F_{is}$ constrasts from $F_{it}$ by calculating the loss of heterozygosity in subpopulations, instead of the total population. The equation for $F_{is}$ is $F_{is}= H_s-H_i/H_s$. The $H_s-H_i$ in the numerator determines the difference between the expected average heterozygosity and observed mean heterozygosity in subpopulations, while the $H_s$ in the denominator contains the expected average herozygosity. The p-values for $F_{is}$ under the FSTAT analyses were based on 800 randomizations and an adjusted nominal at 5%. The frequency of private alleles were determined by drainage and locality using GenAIEx version 6.1 (Peakall and Smouse 2006). A private allele is exclusive to a single population and does not occur in any other population.

The number of clusters, using a Bayesian population assignment protocol, was determined with STRUCTURE 2.3 (Falush *et al.* 2003; Pritchard *et al.* 2000) and GENELAND (Guillot *et al.* 2005). STRUCTURE 2.3 assigns individuals to subpopulations based on analyzing individual genotypes that ultimately formulates the overall population structure (Falush *et al.* 2003; Pritchard *et al.* 2000). Further,

individuals are assigned to a population without the prior need to define a specific population unit. STRUCTURE predicts the number of clusters accurately for diverse scenarios in tests of island model, a contact zone, and hierarchical island model (Evanno *et al.* 2005).

STRUCTURE analyses were run using either an independent allele frequency (Pritchard *et al.* 2000) or correlated allele frequency models (Falush *et al.* 2003). The burn-in period was fixed at $1 \times 10^5$ and a subsequent $1 \times 10^6$ MCMC iterations under the admixture model of ancestry. Genetic clusters (K) underwent 20 iterations for K values (from 1 to 14, which is the maximum number of localities). The best likelihood K value was determined using methods of Pritchard *et al.* (2000) and the $\Delta K$ method outlined by Evanno *et al.* (2005). Alpha value set at default value of 0.025 produced large standard deviations so; the ALPHAPROPSD (standard deviation of the normal distribution describing alpha) was adjusted to 0.05 and 0.1 to promote better mixing by affecting the rate of convergence in the Markov Chain (Falush *et al.* 2003; Pritchard *et al.* 2000).

Even though GENELAND and STRUCTURE can determine the number of populations with multi locus data, GENELAND contrasts with the ability to detect population subdivisions in a spatial framework (Guillot *et al.* 2005). GENELAND version 3.3.0 merges spatial coordinates and genetic data to assign individuals to population subdivisions (Guillot *et al.* 2005). Under 1 to 14 populations, the iterations were fixed at 500,000 with a thinning of 5 for both correlated and uncorrelated frequency. Parameters were set at 100 for x and y with a burn-in of 100.

Bottleneck version 1.2.02 can detect heterozygosity ($H_e$>$H_{eq}$) that may indicate a recent bottleneck event in a population (Cornuet and Luikart 1996). A $H_e$ (measured heterozygosity) and $H_{eq}$ (heterozygosity at equilibrium for a particular locus in a population) would be equal for a nonbottleneck population (Cornuet and Luikart 1996). Mutational models called infinite alleles model (IAM), stepwise mutation model (SMM) and Two Phase Model (TPM) were run in Bottleneck with a 1,000 iterations for each given model (Cornuet and Luikart 1996). IAM accounts for multiple possible mutations in a tandem repeat and the creation of a new allele in a population (Estoup *et al.* 1995). The SMM model is constrained to one addition or subtraction of a tandem repeat and the allele may already exist in the population (Estoup *et al.* 1995). Because of multiple tandem repeats, the IAM model is often preferred over SMM in microsatellite analysis (Estoup *et al.* 1995). Similar to IAM, TPM can detect an excess of hetorozygotes within a population. The proportion of SMM to TPM was set at 70% and 30% for TPM variance. A one-tailed Wilcoxon signed-rank test was used to test for statistical significance. The data was analyzed as a single population and by drainage. Populations in mutation drift-equilibrium were tested to see if the allelic range was full or patchy from a recent bottleneck event (Garza and Williamson 2001).

# CHAPTER 5

# RESULTS

## Microsatellite Loci

Null alleles were present for the loci ECA 6 EPA and ECA 22 EPA when the entire population was analyzed in MICROCHECKER. However, the detection of null alleles varied with a finer analysis based on drainages. ECA 6 EPA had null alleles present for the Cedar River drainage, while ECA 22 EPA had null alleles for the Maquoketa and Turkey river drainages respectively. These loci were not included in any further data analysis except for a single run in STRUCTURE under the independent allele frequency model. According to MICROCHECKER analysis, null alleles were not present in the eight remaining loci.

Assuming one population encompassing 14 localities in GENEPOP, loci ECA 10 EPA (P= 0.0024), ECA 14 EPA (P=0.00), and ECA 44 EPA (P=0.0196) deviated from Hardy-Weinberg equilibrium. However, the remaining loci that met Hardy-Weinberg equilibrium were ECA 24 EPA (P=0.8868), ECA 36 EPA (P=0.2258), ECA 48 EPA (P=0.2963), ECA 46 EPA (P=0.0709) and ECA 71 EPA (P=0.1402). The mean Hardy-Weinberg proportion for each drainage calculated in GENEPOP is listed in Table 3. The average Hardy-Weinberg proportion calculated in ARLEQUIN by drainage was higher than the mean Hardy-Weinberg calculated in GENEPOP: Cedar ($H_e$=0.875378, $H_o$=0.904183), Maquoketa ($H_e$=0.87266, $H_o$=0.848608), Turkey ($H_e$=0.87115, $H_o$ =0.851615), and Upper Iowa rivers ($H_e$=0.876564, $H_o$=0.89411).

GENEPOP results for linkage disequilibrium showed linkage among ECA 14 EPA and ECA 48 EPA (P=0.0067), and ECA 10 and ECA 48 EPA (P=0.04538) accounting for a single population consisting of 14 localities. However, linkage disequilibrium between ECA 10 and ECA 48 EPA, and ECA 14 EPA and ECA 48 EPA were not detected in ARLEQUIN when parameters were set for a single population consisting of 14 localities. Based on the results from ARLEQUIN, ECA 71 EPA and ECA 36 EPA (P=0.029), and ECA 44 EPA and ECA 46 EPA (P=0.025) displayed linkage disequilibrium for the 14 grouped populations. Even though both tests indicated linkage disequilibrium between ECA 44 EPA and ECA 46 EPA, a finer analysis in ARLEQUIN by drainage showed the Upper Iowa (P=0.40743), and Cedar (P=0.45713) rivers as not being significant for linkage disequilibrium between ECA 44 EPA and ECA 46 EPA. In contrast, linkage disequilibrium between ECA 44 EPA and ECA 46 EPA was found for the Turkey (P=0.17683) and Maquoketa (P=0.00208) rivers. Maquoketa (P=0.135568) and Turkey (P=0.08213) rivers were not significant for linkage disequilibrium between ECA 44 EPA and ECA 46 EPA in the GENEPOP analysis. Because of the inconsistent and conflicting values generated from linkage disequilibrium analyses, the eight loci not demonstrating null alleles were all included in subsequent analyses.

Table 3. Allele Size, Number of Alleles, and Hardy-Weinberg Equilibrium based on Drainage. Allele size (bp), observed heterozygosity ($H_o$), expected heterozygosity ($H_e$) for populations in the following drainages: Cedar (C), Maquoketa (M), Turkey (T), and Upper Iowa rivers (UI). An astrick represents a locus failing Hardy-Weinberg equilibrium ($P<0.05$) for GENEPOP analysis of 270 individuals representing 14 localities.

| Locus | Allele Size (bp) | No. Alleles | Ho | | | | He | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | C | M | T | UI | C | M | T | UI |
| *Eca10EPA | 218-282 | 16 | 0.50 | 0.40 | 0.83 | 0.51 | 0.49 | 0.43 | 0.83 | 0.48 |
| *Eca14EPA | 172-256 | 21 | 0.59 | 0.45 | 0.86 | 0.54 | 0.58 | 0.46 | 0.84 | 0.52 |
| Eca24EPA | 163-223 | 16 | 0.60 | 0.42 | 0.86 | 0.46 | 0.55 | 0.43 | 0.86 | 0.51 |
| Eca36EPA | 201-305 | 24 | 0.58 | 0.48 | 0.87 | 0.54 | 0.57 | 0.45 | 0.90 | 0.54 |
| *Eca44EPA | 150-202 | 13 | 0.50 | 0.38 | 0.61 | 0.42 | 0.49 | 0.40 | 0.68 | 0.41 |
| Eca46EPA | 218-326 | 26 | 0.61 | 0.37 | 0.74 | 0.56 | 0.58 | 0.41 | 0.76 | 0.52 |
| Eca48EPA | 165-233 | 17 | 0.55 | 0.43 | 0.82 | 0.53 | 0.55 | 0.43 | 0.85 | 0.53 |
| Eca71EPA | 191-259 | 18 | 0.60 | 0.45 | 0.87 | 0.56 | 0.57 | 0.46 | 0.88 | 0.53 |
| Mean | | | 0.57 | 0.42 | 0.81 | 0.52 | 0.55 | 0.43 | 0.83 | 0.51 |

## Genetic Diversity

Allelic richness, number of alleles, and private alleles showed relatively even distribution across drainages (Table 4 A). The locus that displayed the most polymorphism was ECA 46 EPA with 26 alleles (Table 3). ECA 44 EPA showed the least polymorphic of the eight loci with 13 alleles. The Bear Creek (UI1BC) locality from the Upper Iowa River had five private alleles, while the Maquoketa River locality (MRNL) one mile north of Lamont, Iowa had eight private alleles (Table 4 B). Bear Creek and the Maquoketa River locality in Lamont had the most private alleles in its drainage. East Otter Creek, which is part of the Cedar River Drainage, was the only population that did not contain private alleles. The number of private alleles in each drainage are Cedar (5), Maquoketa (11), Turkey (7), and Upper Iowa (9) rivers.

Table 4. A) Allelic Richness, Allele Frequency, and Private Alleles. Allelic richness (AR), allele frequency per locus (K), and private alleles (Apriv) for populations in the following drainages based on GeneAIEx: Cedar River (CR), Maquoketa(M), Turkey (T), and Upper Iowa (UI) rivers. B) The number of private alleles for each locality. Locality ID corresponds to Table 1. n = the number of individuals.

A)

| Locus | AR | | | | K | | | | Apriv | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | CR | M | T | UI | CR | M | T | UI | CR | M | T | UI |
| Eca10EPA | 11.935 | 13 | 14.536 | 11.912 | 12.0 | 13.0 | 15.0 | 12.0 | 0 | 3 | 0 | 0 |
| Eca14EPA | 15.936 | 14.879 | 14.797 | 17.621 | 16.0 | 15.0 | 15.0 | 18.0 | 0 | 2 | 0 | 3 |
| Eca24EPA | 13.998 | 12.916 | 14.743 | 12.855 | 14.0 | 13.0 | 15.0 | 13.0 | 0 | 2 | 0 | 0 |
| Eca36EPA | 19 | 17.914 | 23.735 | 19.72 | 19.0 | 18.0 | 24.0 | 20.0 | 0 | 0 | 5 | 0 |
| Eca44EPA | 8.936 | 8.919 | 9.966 | 11 | 9.0 | 9.0 | 10.0 | 11.0 | 0 | 0 | 0 | 2 |
| Eca46EPA | 22.84 | 18 | 22 | 17.876 | 23.0 | 18.0 | 22.0 | 18.0 | 3 | 0 | 1 | 2 |
| Eca48EPA | 14.934 | 13.836 | 12.906 | 12.888 | 15.0 | 14.0 | 13.0 | 13.0 | 2 | 1 | 0 | 2 |
| Eca71EPA | 14.936 | 13.938 | 16.883 | 12.878 | 15.0 | 14.0 | 17.0 | 13.0 | 0 | 3 | 1 | 0 |
| average | 15.314 | 14.1753 | 16.1958 | 14.5938 | 15.4 | 14.3 | 16.4 | 14.8 | 0.63 | 1.4 | 0.9 | 1.1 |

B)

| Drainage | ID | n | Apriv |
|---|---|---|---|
| Upper Iowa | UI1BC | 10 | 5 |
| | UI2HR | 8 | 1 |
| | UI3BG | 20 | 2 |
| | UI4M | 21 | 1 |
| Turkey | TK1WC | 30 | 3 |
| | TK2SC | 30 | 1 |
| | TK3FA | 25 | 2 |
| | VTVS | 12 | 1 |
| Maquoketa | MRBP | 11 | 2 |
| | MRNM | 15 | 2 |
| | MRNL | 25 | 8 |
| Cedar | OCNA | 30 | 3 |
| | RCSO | 12 | 2 |
| | EOC | 22 | 0 |

$F_{st}$ values suggest very little divergence and a lack of population subdivisions (Table 5; Table 6). For GENEPOP analyses by drainage, the $F_{st}$ values ranged from 0.0087 through 0.0198 for each of the eight loci. $F_{st}$ values among localities within drainages even showed a lack of structure (Table 6). Localities within drainages had the following mean for $F_{st}$ values representing the eight loci, Cedar (0.0021), Maquoketa (0.0008), Turkey (0.0106), and (0.0079) Upper Iowa rivers. $F_{it}$ tests for an abatement in heterozygotes within an individual in relation to the overall population. Values of $F_{it}$ were all close to zero with four populations representing each drainage (Table 5).

Table 5. $F_{is}$, $F_{st}$, and $F_{it}$ values for each Locus generated in FSTAT.

| Locus | F(is) | F(st) | F(it) |
|---|---|---|---|
| ECA 10 EPA | -0.0002 | 0.0145 | 0.0142 |
| ECA 14 EPA | -0.0182 | 0.0198 | 0.002 |
| ECA 24 EPA | 0.0043 | 0.0145 | 0.0187 |
| ECA 36 EPA | -0.0079 | 0.0087 | 0.0009 |
| ECA 44 EPA | 0.0353 | 0.004 | 0.0392 |
| ECA 46 EPA | -0.0004 | 0.0089 | 0.0085 |
| ECA 48 EPA | 0.0174 | 0.0122 | 0.0294 |
| ECA 71 EPA | -0.0194 | 0.0097 | -0.0095 |
| All: | 0.0005 | 0.0117 | 0.0122 |

Table 6. $F_{st}$ values from 14 Localities. Sampling locations codes are in Table 1

| | UI4M | UI3BG | UI1BC | UI2HR | TK1WC | TK4FA | TK2SC | VTVS | MRNL | MRNM | MRBP | OCNA | RCSO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UI3BG | 0.003 | | | | | | | | | | | | |
| UI1BC | 0.006 | 0.009 | | | | | | | | | | | |
| UI2HR | 0.023 | 0.011 | 0.012 | | | | | | | | | | |
| TK1WC | 0.022 | 0.015 | 0.016 | 0.031 | | | | | | | | | |
| TK3FA | 0.011 | 0.008 | 0.014 | 0.026 | 0.014 | | | | | | | | |
| TK2SC | 0.005 | 0.005 | 0.01 | 0.025 | 0.013 | 0.002 | | | | | | | |
| VTVS | 0.009 | 0.005 | 0.008 | 0.011 | 0.019 | 0.012 | 0.006 | | | | | | |
| MRNL | 0.02 | 0.02 | 0.017 | 0.032 | 0.017 | 0.014 | 0.009 | 0.02 | | | | | |
| MRNM | 0.015 | 0.02 | 0.014 | 0.024 | 0.027 | 0.011 | 0.013 | 0.014 | -0.002 | | | | |
| MRBP | 0.02 | 0.026 | 0.019 | 0.007 | 0.022 | 0.024 | 0.016 | 0.01 | 0.005 | -0.0007 | | | |
| OCNA | 0.01 | 0.009 | 0.006 | 0.024 | 0.022 | 0.004 | 0.012 | 0.012 | 0.023 | 0.0111 | 0.024 | | |
| RCSO | 0.019 | 0.008 | 0.006 | 0.014 | 0.021 | 0.006 | 0.016 | 0.016 | 0.022 | 0.008 | 0.019 | -0.004 | |
| EOC | 0.01 | 0.013 | 0.017 | 0.017 | 0.031 | 0.016 | 0.015 | 0.016 | 0.033 | 0.0148 | 0.018 | 0.0057 | 0.0006 |

## Population Structure

A single cluster was resolved in the STRUCTURE analyses based on a Bayesian admixture approach. Evaluation of [Pr(X/K)] showed the most likely number of clusters encompasses a single population (K=1) (Figure 8). Conversely, ΔK had a value of two clusters. However, this method cannot distinguish the difference between two populations or a single population (Evanno *et al.* 2005). Because of the inability of ΔK to find a single population, Evanno *et al.* (2005) recommends running likelihood [Pr(X/K)] to complement ΔK. When the two null alleles were incorporated into the correlated allele frequency at 0.1 ALPHAPROPSD, the ΔK=2 and the likelihood [Pr(X/K)] also suggested a single, large population (Figure 8 C). However, excluding the

null alleles from the correlated allele frequency model with admixture yielded a very low peak for ΔK (Figure 8 D). Parameters for ALPHAPROPSD implemented at 0.025, 0.05, and 0.1 under the independent allele frequency model indicate ΔK=2, and a likelihood [Pr(X/K)] value of one cluster (Figure 8 A and B).

GENELAND confirmed the results from STRUCTURE that there is a single population. By incorporating spatial data of UTM coordinates into genetic data, GENELAND confirmed a single population of rainbow darters in northeast Iowa. The probability for a single cluster was greater than 50%, while the chances of a multiple clusters existing were all below 20% (Figure 9).

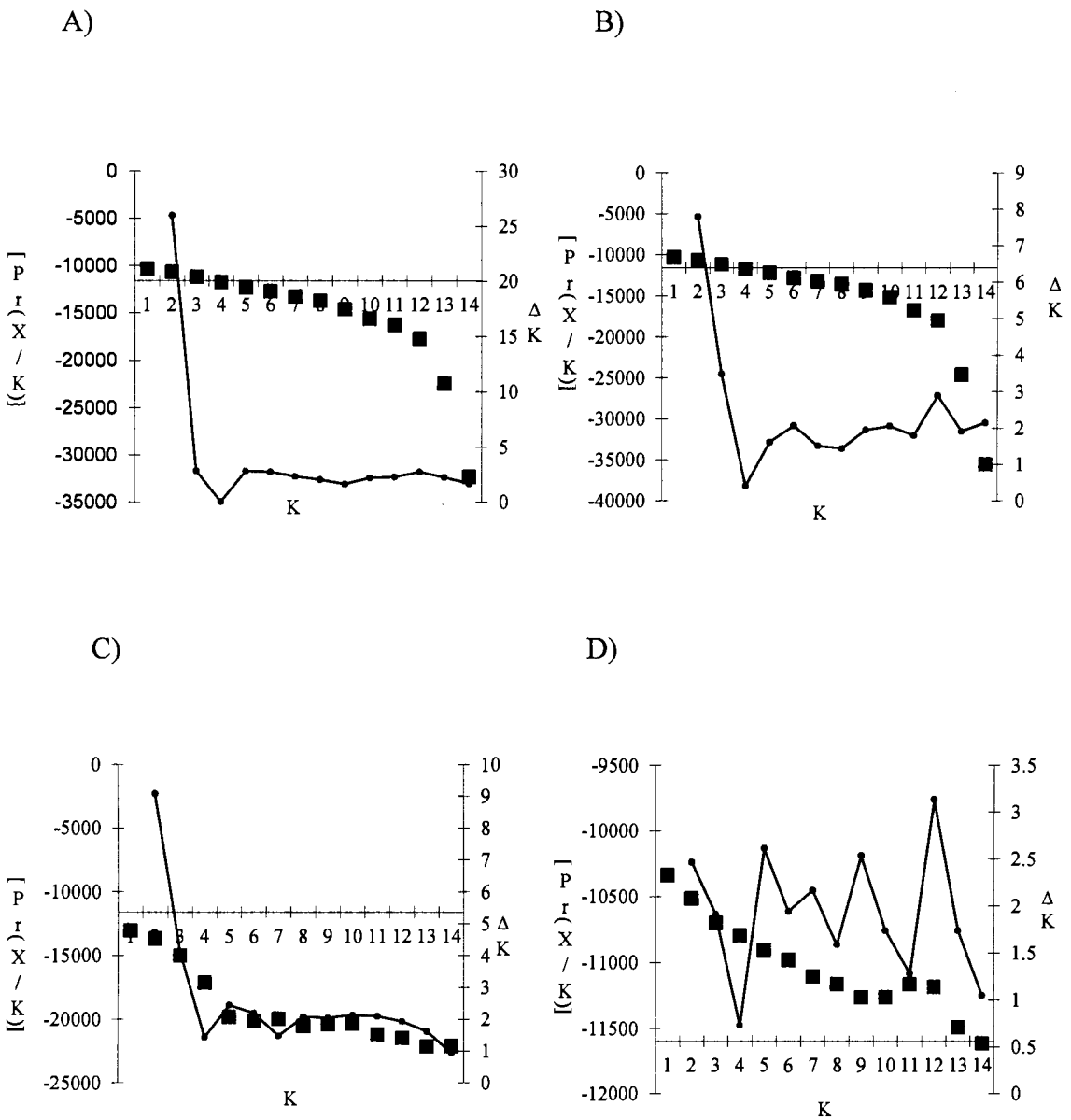


Figure 8. Plot of ΔK and [Pr(X/K)] values as calculated by STRUCTURE. The solid line for K indicates the ΔK value and the black square represents [Pr(X/K)] values for K=1-14 as calculated in STRUCTURE. A) Allele frequency was assumed to be independent with a alphpropsd set at 0.1. B) Allele frequency were assumed to be independent with a alphpropsd of 0.05. C) Allele frequency was assumed to be correlated with a alphpropsd of 0.1. with null alleles D) Allele frequency was assumed to be correlated with a alphpropsd of 0.1. without null alleles.

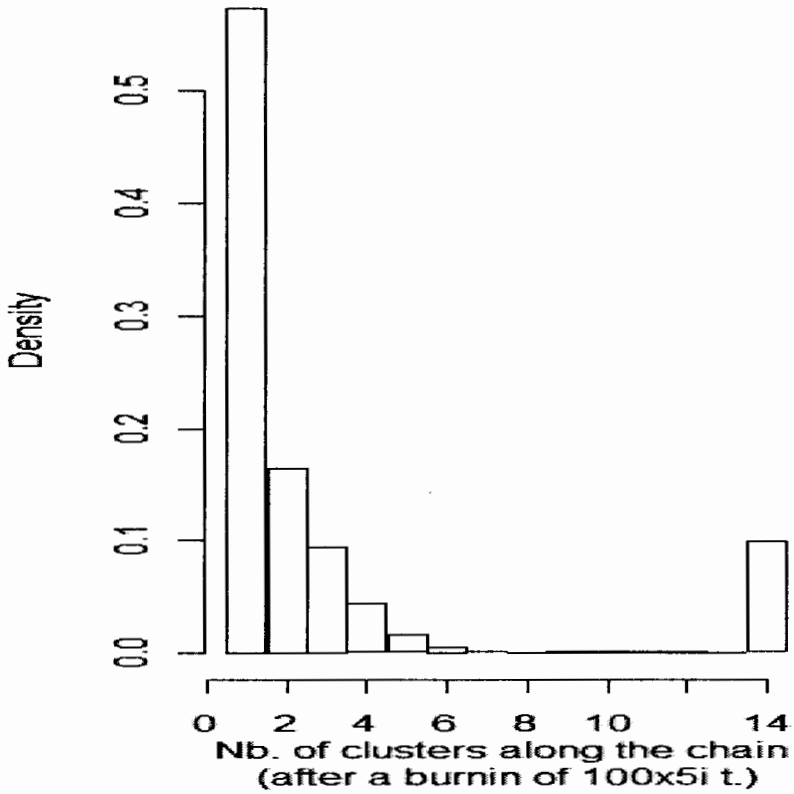


Figure 9. A Histogram of the Number of Clusters generated by GENELAND. The bar with the highest density signifies the most likely number of clusters, which is one cluster.

## Bottleneck

Results of IAM and TPM indicate an excess of heterozygotes and a recent bottleneck within each drainage. Infinite alleles model (IAM) showed excess heterozygosity was pronounced in the Cedar (8), Maquoketa (7), Turkey (7), and Upper Iowa (8) river drainages for each locus. The p-values for each drainage under the IAM was Cedar (P=0.00195), Maquoketa (P=0.00391), Turkey (P=0.00391), and Upper Iowa (P=0.00195) river drainages. TPM shared the same pattern with the Cedar (5), Maquoketa (7), Turkey (6), and Upper Iowa (5) river drainages in regards to for all eight

loci. The following p-values for each drainage under TPM was Cedar (P=0.125), Maquoketa (P=0.09766), Turkey (P=0.15625), and Upper Iowa (P=0.09766) river drainages. SMM, which is not a preferred model for microsatellite analysis (Estoup *et al.* 1995), had an excess of homozygotes for most drainages. The excess of heterozygosity under SMM was Cedar (3), Maquoketa (2), Turkey (1), and Upper Iowa (3) for each locus. Allele frequencies for the four drainages had an L-shaped distribution.

# CHAPTER 6

# DISCUSSION

The analyses of eight microsatellite loci from 14 localities of *Etheostoma caeruleum* detected a single, genetic population in northeast Iowa. The Bayesian clustering method of STRUCTURE and GENELAND of loci not exhibiting null alleles indicated a single population (Figure 8 A and B; Figure 9). The low $F_{st}$ values imply a lack of population structure and divergence (Table 5; Table 6). There is no evidence that geological regions or anthropogenic barriers isolate local populations.

There are two potential explanations why the rainbow darter exhibited no population genetic structure. First, the rainbow darter could be actually highly migratory in northeast Iowa resulting in a uniform population structure. However, the high migration scenario does not seem plausible considering the life history of the rainbow darter (Page 2000). The rainbow darter is thought to be constrained to a few kilometers during its lifetime (Page 2000). In addition, genetic studies by Ray *et al.* (2008) and Haponski *et al.* (2009) revealed low levels of gene flow in the rainbow darter.

The second explanation for the absence of population subdivision is that the genetic signature is a product of historical processes. The characteristics of a microsatellite can influence the perceived genetic structure of a population. This is attributed to the deep coalescence time of microsatellite alleles, which is dictated by effective population size, and the assumption that the mutation rate is independent of the marker (Zink and Barrowclough 2008). The mutation rate of a microsatellite can impact the number of alleles at a particular locus but does not have ramifications in regards to

the impact on a longer depth in time as nuclear DNA sequences (Zink and Barrowclough 2008). Consequently, one can mistakenly deduce that microsatellites failing to detect population structure are a product of bias in gene flow. Instead, the lack of population structure may be attributed to a greater lag time (Zink and Barrowclough 2008).

These results support the alternate hypothesis that populations of *Etheostoma caeruleum* recently colonized northeast Iowa. This is consistent with what has been observed in other organisms. Aquatic organisms in southeastern regions of North America were shown to display more population substructure when compared to population substructure of aquatic organisms in previously glaciated regions in the North (Avise 2000). A newly established population in a common area lacks high population genetic structure and shows signs of past bottlenecks. Ray *et al.* (2006) documented this pattern of rainbow darter expansion in to previously glaciated regions in response to the Wisconsinan glacial retreat using DNA sequence data. They found that the east-central rainbow darter clade consisting of individuals from the Eastern Highlands and streams located north of the Ohio River did not display structure based on mtDNA analysis. Interestingly, some of the Eastern Highlands (portions of Kentucky, Tennessee, and southern Ohio) shared a similar history with the Paleozoic Plateau and Iowan Surface in that they were circumnavigated by the Wisconsinan glacial advance (Figure 1). The absence of population structure in the east-central rainbow darter populations was attributed to either recent recolonization or gene flow (Ray *et al.* 2006).

Microsatellite Loci

Null alleles were detected for the loci ECA 6 EPA and ECA 22 EPA. Because of the detection of null alleles, ECA 6 EPA and ECA 22 EPA were excluded from further data analyses. Even though there were eight loci remaining, the number of loci was sufficient enough to determine the population genetic structure (Ruzzante 1998). Haponski *et al.* (2009) found null alleles present for ECA 49 EPA in Big Darby Creek and Little Miami River in Ohio. Markers ECA 48 EPA from the Blanchard River, ECA 11 EPA from the Big Darby Creek, and ECA 14 EPA from the Little Miami River also showed evidence of null alleles (Haponski *et al.* 2009). The loci were incorporated into their data analyses because the null alleles were in Hardy-Weinberg equilibrium (Haponski *et al.* 2009). Tonnis (2006) reported one null allele for ECA 2 EPA and none were detected in the remaining sampled loci. Null alleles can occur for several reasons such as: a primer not annealing properly to a diverged DNA sequence as a result of mutations, shorter alleles amplifying more efficiently than larger alleles, and DNA template quantity and quality (Dakin and Avise 2004). Consequently, one can be duped into interpreting a heterozygote individual as homozygote individual (Dakin and Avise 2004). Mutations, different populations, and anomalies in the amplification process may explain the contrasting results for ECA 48 EPA and ECA 14 EPA being a null allele in the Ohio population and the absence of a null allele for ECA 48 EPA and ECA 14 EPA in Iowa populations.

Most of the loci used by Haponski *et al.* (2009) were in Hardy-Weinberg equilibrium and displayed no signs of linkage equilibrium with the exception of ECA 11

EPA in the Blanchard River population. However, the results for Hardy-Weinberg equilibrium and linkage equilibrium of Haponski *et al.* (2009) study contrasted with results obtained from rainbow darter populations in this study. ECA 14 EPA was found to be in Hardy-Weinberg equilibrium in sampled Ohio populations in Haponski *et al.* (2009), while ECA 14 EPA failed one of the Hardy-Weinberg equilibrium tests for this study. Haponski *et al.* (2009) results indicated that ECA 10 EPA, ECA 14 EPA, ECA 46 EPA, and ECA 71 EPA met Hardy-Weinberg equilibrium. The statistical significance of Hardy-Weinberg and linkage disequilibrium varied by the type of analyses carried out and even by how fine a population was defined in this study. Because of the different models employed and population definition criteria yielding contrasting results, the eight loci passed the tests for Hardy-Weinberg equilibrium and linkage equilibrium. Therefore, all loci were used in subsequent analyses.

## Population Genetic Structure

Even though the Bayesian clustering method of STRUCTURE and GENELAND estimated a single population, there were a few populations rich in private alleles. The averages for the number of alleles and allelic richness were almost uniform for the Cedar, Maquoketa, Turkey, and Upper Iowa river drainages (Table 4 A). However, the number of private alleles varied between the four drainages and localities (Table 4 A; Table 4 B). Bear Creek (UI1BC), which flows into the Upper Iowa River, had five private alleles and the remaining three localities distributed upstream along the Upper Iowa River had four private alleles (Figure 7; Table 4 B). Ten rainbow darters were collected from Bear Creek, while some of the localities upstream had more fish collected with 8, 20, and 21

individuals respectively (Table 4 B). Sampling error can potentially influence the frequency of private alleles quantified in a given population with larger populations generally containing more private alleles (Kalinowski 2004). The combination of seine and electroshocking employed in this study is considered one of the most effective methods for evaluating species number (Bayley *et al.* 1989). In addition, thorough sampling of a genetically similar population can decrease the frequency of private alleles (Kalinowski 2004). Haponski *et al.* (2007) found *Percina caprodes* located below a low-head dam had unique haplotypes and a lack of gene flow upstream of a barrier. Perhaps the difficulty of upstream migration can explain why the Bear Creek population in this study had more private alleles than the upstream localities. Furthermore, the Upper Iowa River flows through the Paleozoic Plateau that has a steeper gradient than other regions in Iowa. Elevation was shown to influence the population genetic structure of rainbow trout, *Oncorhynchus mykiss*, in areas impacted by the Wisconsinan glacial advance (Tamkee *et al.* 2010).

The MRNL locality from the Maquoketa River had eight private alleles, while the other populations along the Maquoketa River had a combined three private alleles (Figure 7; Table 4 B). The presence of eight private alleles in the MRNL locality may suggest some genetic structure and a genetically distinct population. A dam greater than 4 meters separates MRNL from the other two Maquoketa River localities, which may explain the diversity of private alleles upstream. The absence of private alleles in the East Otter Creek population, which flows into the Cedar River, can be attributed to either high connectivity or additional evidence of recent colonization. The scenario for high

present-day migration between the two sampled Cedar River drainage populations and the East Otter Creek population seems dubious considering that the Cedar River averages an impoundment every 15 miles (Fields 2008). Future instream studies of rainbow darters movement in the rivers of northeast Iowa are necessary to determine if these hypotheses are supported.

Colonization after the Wisconsinan Retreat

The single population of rainbow darters encompasses both the Iowan Surface and Paleozoic Plateau of northeast Iowa. The lack of population structure suggests that the rainbow darter recently colonized northeast Iowa following the Wisconsinan glacial episode. In periods of intense soil erosion in northeast Iowa during the Wisconsinan episode (Iannicelli 2010), heavy erosion increased the sediment loads in Iowa tributaries (Bettis III and Autin 1997). Considering the rainbow darter life history as a specialist that prefers habitat consisting of cobbles and gravel (Heithaus and Laushman 1997), the heavy erosion of the late Wisconsinan episode could have altered the habitat dynamics to the point where *Etheostoma caeruleum* was unable to survive in northeast Iowa. Local extinctions driven by glacial activity are a product of dramatic changes in habitat conditions, which force populations to seek out new refuge or refugia (Hewitt 1996).

Even though an unglaciated northern pocket may be available to some species, some populations of organisms were only able to persist in the Ozarkian refugium (Burr and Page 1986) such as the Ozark minnow (Berendzen *et al.* 2010). The Ozark minnow displayed a similar pattern to the rainbow darter by relocating to a southern refugium in response to glacial activity. Ozark minnows were unable to survive in the Paleozoic

Plateau during the cold periods of the Wisconsinan episode and were forced to seek out southern refuge in the Northern Ozark Plateau (Berendzen *et al.* 2010). As the Wisconsin glacial advance retreated, the Ozark minnows recolonized the upper Mississippi River region, e.g. Paleozoic Plateau, from the Northern Ozark refugium (Berendzen *et al.* 2010). The establishment of new drainage networks from glacial activity enabled fish to colonize new areas and to expand their range (Berendzen *et al.* 2003). Factors that influence the prosperity of a founding population include the ability of a species to adapt in a new environment, habitat distribution, and climate oscillation modifying habitat (Hewitt 2004). Populations of Ozark minnow were extirpated in southern Iowa and northern Missouri in response to increased sediment load during glacial activity, which left the Northern Ozark and upper Mississippi River populations disjunct from each other (Berendzen *et al.* 2010).

Interestingly, the rainbow darter displays the same disjunct pattern of not being found in southern Iowa and northern Missouri (Lee *et al.* 1981). They occur in the Paleozoic Plateau of the upper Mississippi River and the Northern Ozarks of southern Missouri and are absent from southern Iowa and northern Missouri (Lee *et al.* 1981). Despite the rainbow darter populations of the Ozark Highlands and northeast Iowa being separated geographically from each other, shared haplotypes between these regions offer insight into the origins of the rainbow darter population in Iowa (Figure 6) (Ray *et al.* 2006). The Osage River, which flows through Missouri, forms a clade that is sister to the upper Midwest clade (Ray *et al.* 2006). This relationship implies the Osage River was one of the routes used by the rainbow darters to colonize northeast Iowa. There is also a

possibility that the rainbow darter colonized northeast Iowa from Wisconsin because they share a common haplotype (Figure 6) (Ray *et al.*2006). However, rainbow darter colonization from Wisconsin seems unlikely because intense erosion could have created unsuitable habitat in the Driftless Area (Iannicelli 2010). Both the rainbow darter and Ozark minnow likely colonized Iowa from a refuge in the Northern Ozark Plateau.

Founder Effect

A repercussion of a population colonizing a previously unglaciated region elicits the founder effect. A founder effect occurs when a new population containing a few individuals colonizes a new area from a more genetically rich source population. One of the consequences of the founder effect is a bottleneck where a small colonizing population is prone to changes in allele frequency and erosion of genetic diversity (Nei *et al.* 1975). Two types of evidence that support a recent bottleneck in rainbow darter populations are excess heterozygotes and low frequency of private alleles. Four drainages illustrate excess heterozygosity present in either seven or eight markers. The implications of excess heterozygosity entails that a population may have undergone a recent bottleneck (Cornuet and Luikart 1996) and recent bottlenecks were found to have an affiliation with colonization (Pascual *et al.* 2001). A study of a land snail, *Theba pisana,* which invaded and colonized Australia, contained a few private alleles. This implied the population underwent a recent bottleneck (Johnson 1988). A deficiency in private alleles across eight loci in the Cedar, Maquoketa, Turkey, and Upper Iowa rivers could be evidence that rainbow darter has undergone a recent bottleneck but there is no population from a southern refugium in this study to compare private allele frequencies

to. The four sampled drainages in Iowa had a total of 32 private alleles. Ohio rainbow darter populations had 21 private alleles in the Lake Erie catchment and 25 private alleles in the Ohio River catchment for eight loci (Haponski *et al.* 2009). Five out of eight loci from Haponski *et al.* (2009) were the same loci employed in this study.

Unlike the dearth of genetic diversity found in newly established populations, a refugium during glacial advances tends to house more genetic richness (Lee-Yaw *et al.* 2008; Demastes *et al.* 2007; Rowe *et al.* 2004). A colonizing population of rainbow darter in the Lake Erie catchment had less genetic diversity than the refuge population in the Ohio River catchment (Haponski *et al.* 2009). Further, mtDNA from Haponski *et al.* (2009) results suggest that the newly colonized rainbow darter population in the Lake Erie catchment experienced the founder effect. Ray *et al.* (2006) documented the same pattern with more genetic diversity in refugial areas than in regions glaciated by the Pleistocene advances. The Northern Ozarks-upper Midwest clade combined had a nucleotide diversity of 0.017±0.0009, while the sampled unglaciated region showed richer nucleotide diversity of 0.035±0.020 (Ray *et al.* 2006). Four out of the five loci from the Iowa populations showed more polymorphism when compared to the same loci from rainbow darter populations in Ohio. Unfortunately this study lacks samples from southern populations to compare the colonizing northeast Iowa rainbow darter population to.

## Conservation Implications

Because the rainbow darter populations in Iowa appear to exist as a single, genetic population, the population should be managed as a whole unit. There is no evidence that

low head dams and impoundments currently influence the population structure of the rainbow darter. Other studies on log perch, (Haponski *et al.* 2007) and rostrum dace (Blanchet *et al.* 2010) indicated that anthropogenic structures indeed impact population genetic structure. Even though populations of rainbow darters are separated by anthropogenic barriers, their genetic integrity seems to be fine considering the excess of heterozygotes present in all four drainages. The current genetic signature is a relic of historical processes because the recently established anthropogenic structures have not persisted long enough to provide ample time for subpopulations to display structure. A population of rainbow darters in the MRNL locality harbored some structure with eight private alleles although the population analyses indicated the presence of a single, genetic population. The single population may diverge into subpopulations because of the life history of the rainbow darter, habitat specialist, and anthropogenic structures sequestering gene flow between populations. Therefore, the outcome of abated gene flow may lead to local adaptations in subpopulations. Since populations of rainbow darter in northeast Iowa occur as a single, genetic population, the conservation strategy should focus on managing the population as a single unit. The single, genetic population of rainbow darters in Iowa likely encompasses Minnesota and Wisconsin but more research is required to confirm the extent of genetic similarity.

# CONCLUSION

The recent bottleneck and single, genetic population suggests that the rainbow darter recently colonized northeast Iowa after the retreat of the Wisconsinan glacial episode. The Northern Ozark Plateau may have served as a refugium that enabled the rainbow darter to colonize Iowa. Gene flow may be limited upstream in the Upper Iowa drainage as supported by private alleles being more prevalent in downstream populations. However, private alleles were richer upstream when compared to the downstream site of the Maquoketa River. Despite undergoing a recent bottleneck, the current populations rainbow darter do not display evidence of inbreeding. Historical processes reflect the current genetic signature of the rainbow darter more so than anthropogenic barriers. Future rainbow darter population genetic studies may want to employ microsatellites and mtDNA in regions not glaciated by the Quaternary glacial advance, e.g., Northern Ozark Plateau.

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